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Seminar: Applied Corporate Finance

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Valuation of Visa

Rethinking Cost of Equity for Visa with Intertemporal Asset Pricing Models

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Decleration of Authorship

Abstract, introduction, conclusion and bibliography review have been written jointly by the authors.

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Abstract

With rising interest rates, the cost of capital for firms has increased at a rapid pace in the last 18 months causing the share price of especially growth stocks to decline. In this context, this paper estimates the fundamental value of Visa, Inc, with an emphasis on the cost of capital. This is achieved by conducting an analysis of Visa's strategic position and financial statements used to forecast Visa's free cash flows and applying a discounted cash flow model. For the cost of capital, this paper studies if the cost of equity provided by a CAPM model is appropriate with Visa's strong competitive advantages. This paper applies the alternative Good Beta, Bad Beta model which suggests different risk premiums for cash flow and discount factor risk. With the CAPM, the intrinsic value of a Visa share is found to be 216 USD, 6 % lower than the market price on 25-04-2023 of 230 USD. On the other hand, the lower cost of equity suggested by the good beta bad beta models results in an intrinsic of 299 USD, 30 % higher than Visa's share price. While this suggests the market uses CAPM to estimate Visa's cost of equity, the good beta, bad beta regression provides a significantly higher R^2 meaning it might be a better fit to explain Visa's realised returns. Finally, through a sensitivity analysis, this paper concludes that all results are highly sensitive to subjective data and method choices in both forecasting Visa's cash flows and estimating the cost of capital.

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Introduction

Main Research Question

What is the fundamental value of common shareholder equity in Visa Inc?

Subquestions

1. How will Visa's free cash flows develop in the next 10 years and beyond? What are the key macro- and micro-indicators driving Visa's financial results?
2. What is the intrinsic value of Visa's shares using both the CAPM and other relevant cost of equity models and which model is better at explaining Visa's realised returns?
3. Is Visa's current share price more in line with a cost of equity predicted by the CAPM or other relevant cost of equity models? How sensitive are the results to changes in data choices for both the projected cash flows and the cost of capital?

Introduction

Since March 2022, the Federal Reserve's (FED's) funds rate has been raised at 10 consecutive meetings going from an effective rate of 0.08 % to 5.08 % [1]. This level has not been reached since 2007 and the increases have come at the fastest pace since the 1980s. In the same period, the CRSP US Total Market Index (TMI) has decreased by 10 % attributed to the increasing rates by market analysts [1]. The interest hikes from the FED affect the stock market as two components are required for pricing any asset: the future cash flows and the appropriate discount rate reflecting the opportunity cost [2]. Higher interest rates increase discount rates as the market pays a higher return on new investments. This means all other investments are relatively worse and stock prices drop.

In this context, we find it interesting to estimate the fundamental value of shares in Visa Inc. Visa is highly exposed to interest rate hikes as a large amount of Visa's fundamental value is based on future growth and increasing discount factors thereby significantly decrease Visa's present value. On the other hand, Visa has the potential for strong long-term cash flows from its dominating market position and strong competitive advantages which is attractive to investors. Visa is described as a "tax on electronic private consumption" which may have short-term fluctuations, but as long as economies continue to grow, so will Visa's lucrative long-term revenue stream. Some investors with a long-term investment horizon such as Warren Buffet¹ express the view that high certainty of long-term future cash flows stemming from competitive advantages is not reflected sufficiently when academics risk-adjust cash flows.

"No formula in finance tells you the moat [competitive advantage of a company] is 28 ft wide and 16 ft deep. That's what drives academics crazy. They can compute standard deviations and betas, but they can't understand moats. Maybe I'm being too hard on the academics."
- Warren Buffet (1999)²

One financial theory which incorporates Buffet's view is the Good Beta, Bad Beta (GBBB) model by Campbell & Vuolteenaho (2004) [4] built on theory from the intertemporal CAPM (ICAPM). Here, the traditional CAPM market risk beta is split into a cash flow beta and a discount factor beta. The representative investor requires a lower expected return to take on discount factor beta as this might reduce the value of current investments but increases future expected returns. On the other hand, the investor requires a high expected return to take on cash flow market risk as this only reduces future consumption.

¹According to most recent SEC filings, Warren Buffet is currently a Visa shareholder, [3]

²At BRK 1999 annual meeting

This paper estimates the intrinsic value of shares in Visa with all information available as of 25-04-2023³ to be 216 USD with the CAPM compared to 299 USD with the Good Beta, Bad Beta model. The higher value is due to the estimated cost of equity being 1.25 %-points lower in the Good Beta, Bad Beta model which is found to be overall better at explaining the realised returns of Visa. This suggests that Good Beta, Bad Beta could be a superior model for estimating Visa's true cost of equity. On the other hand, Visa's current share price of 230 USD⁴ is in line with our valuation using the CAPM. This suggests that the market values Visa's shares using the CAPM. However, all results are highly sensitive to different assumptions in both estimated future cash flows and the cost of capital.

The rest of the paper is as follows: Section 2 is an introduction to Visa and the payment ecosystem. Section 3 is a short strategic analysis and Section 4 is a brief financial statement analysis which are both used in Section 5 to forecast Visa's financials for the next 10 years. Section 6 determines the cost of capital with emphasis on the cost of equity. Section 7 is the valuation and sensitivity analysis of the results.

³The exact day their 2023Q2 report was released

⁴At: 25-04-2023

Company Presentation

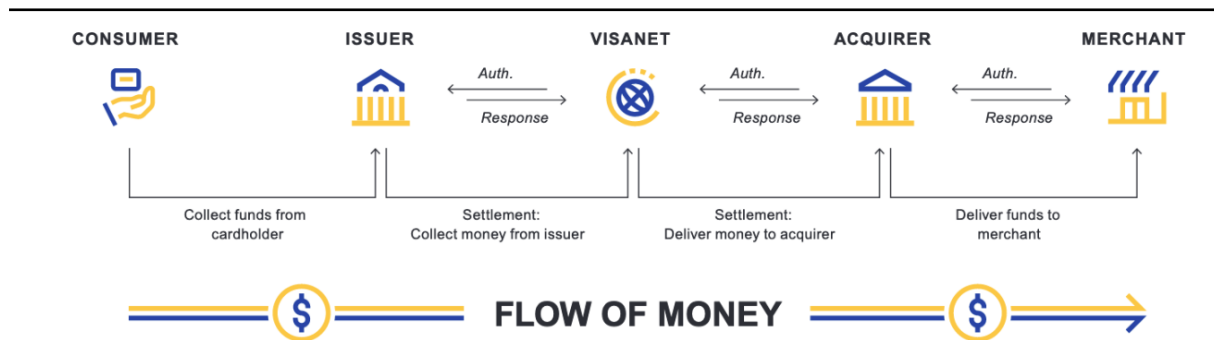
2.1 Company Profile

Visa, Inc. is an American company that facilitates electronic fund transfers through its payment network, known as VisaNet [5]. In 2022, VisaNet processed 192.5 billion transactions and had a payment volume of \$11.6 trillion split almost evenly between the US and the rest of the world [5]. Visa is generating the majority of its revenue by taking a combination of a flat (\$) and variable (%) fee for facilitating transfers. For this reason, Visa has been informally described as "a toll company for electronic payments" or as "a tax on electronic private consumption" [6]. As emphasised by Visa [5], they do not issue credit cards, extend credit or set rates and fees for account holders of their products. As a result, Visa does not bear any credit risk.

Core business: Consumer-to-Business Card Payments

Visa's core business is facilitating Consumer-to-Business (C2B) payments for their credit, debit and pre-paid cards. The flow of payments has four parties: *i*) the consumer, *ii*) the bank of the consumer (issuer), *iii*) the bank of the merchant (acquirer) and *iv*) the merchant.

Figure 2.1: The four-party model



Source: Visa's Annual Report 2022 [5]

VisaNet facilitates efficient communication between the issuer and acquirer, which allows for payments to be cleared, settled, and authorized. In addition, VisaNet also provides extensive security checks for both parties¹. For instance, communication through VisaNet ensures the consumer has the necessary funds to make the payment, but it also rejects payments deemed to be fraudulent [7].

¹Under regulatory scrutiny, Visa has argued their extensive payment security processes such as allowing consumers to later dispute payments are a main factor when considering what societal value Visa contributes

New flows: expanding into new business segments

In addition to their core C2B business, Visa also aims to grow by moving into facilitating person-to-person (P2P), business-to-consumer (B2C), business-to-business (B2B), and government-to-consumer (G2C) payments [5]. The most successful and mature initiatives (measured by network payment volume) are 'Visa Direct' and 'Visa Business Solutions'. Visa Direct allows direct transactions between bank accounts targeting mostly P2P (similar to Denmark's MobilePay). Visa Business Solutions targets merchants to pay using VisaNet and thereby is an expansion into the B2B payment solutions space.

2.2 Payment Solutions Ecosystem

The payment ecosystem is complex and consists of various entities playing different roles besides payment networks such as Visa. Here, we give a basic introduction that is required as background knowledge for the strategic analysis and the assessment of Visa's true peers². In addition to the already described payment networks, the three parts of the ecosystem we present³ are Acquirers, Issuers and Gateways [7]. The issuer is the bank of the consumer issuing the card using a specific payment network such as JPMorgan Chase or Danske Bank issuing a Visa card. An acquirer, also called the acquiring bank, is the bank connecting the merchant to the payment network. This service can also be provided by a bank such as JPMorgan Chase or Danske Bank. However, there are niche acquiring banks focusing on online payments such as Ayden and Fiserv [7]. The final role covered here, Gateways, connects online webpage payments to an acquiring bank with key players being PayPal⁴, Square and Ayden (Ayden is both an Acquirer and a Gateway).

Figure 2.2: The Payment Ecosystem



Source: Own Analysis

²For instance Apple Pay and Mobile Pay are not Visa's peers as users type in their card information thereby using VisaNet

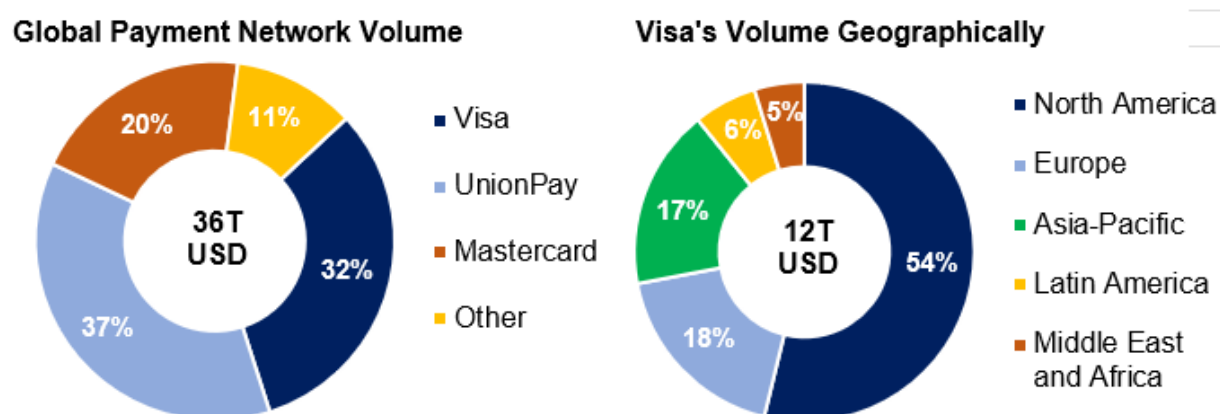
³One key omitted part is ISOs which are very similar to acquirers but payment solutions to merchants

⁴Called a Payment Service Provider (PSP)

Because of this division in the payment ecosystem, Visa's only core peers are other Payment Networks such as MasterCard, American Express and Discover. When we refer to Visa's market, we therefore only mean the payment network market. However, the big niche players in other parts of the ecosystem such as Fiserv, PayPal, Ayden and Square can be considered superficial peers as their financial performance also heavily depends on the payment volume in the ecosystem. In any transaction, payment networks are paid what is known as the assessment fee. For Visa, this is on average 0.14 % of the transaction volume plus a flat fee of \$0.0195, but it varies greatly across card types and geography [8]. The payment to the issuing bank is called the interchange fee which is much larger, for Visa between 1.15 % and 2.40 % [8]. The total of these are informally referred to as swipe fees.

The Payment Network market is dominated by a few players with large market shares. UnionPay is the market leader with 37 %, but has most of its presence in China. In the rest of the world, Visa is the market leader with 32 % of global payment volume and almost half outside the US. Visa especially has a significant presence in Europe and the Asia-Pacific.

Figure 2.3: Payment Network Volume Split on Network and Visa's Geographies: 2022

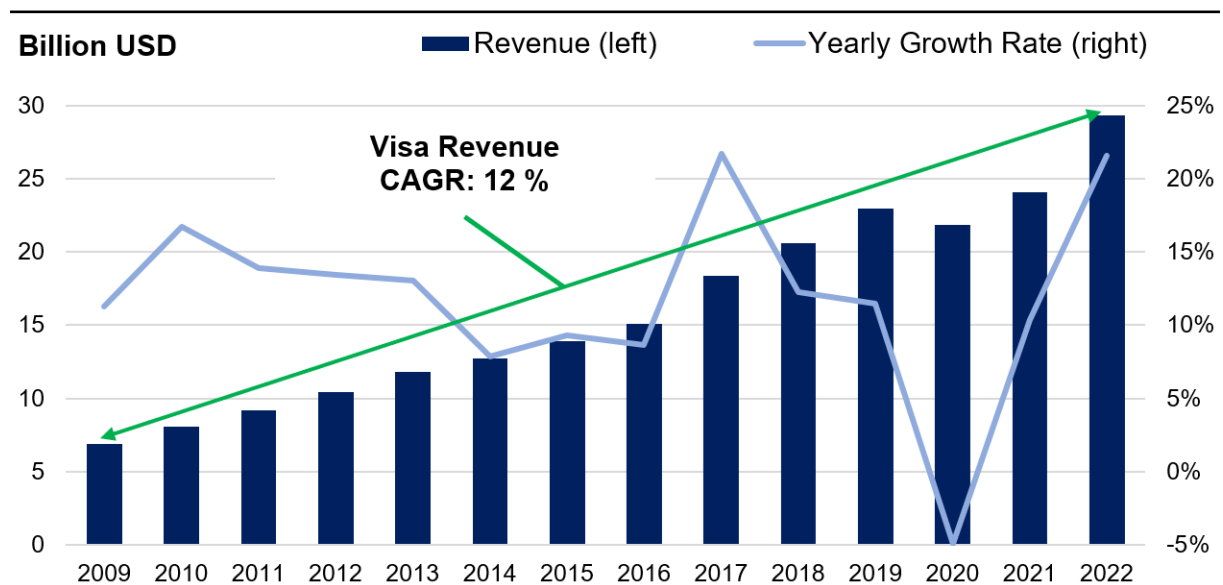


Notes: Domestic Payment Networks without a global presence are excluded
Source: Nilson Report 1233 [9] & Visa's Operations Report 22 [10] & 2022 [5]

2.3 Financial Overview

Visa's reported revenue for 2022 was 29 billion [5], a 22 % increase from 2021. Since 2009, Visa's revenue has grown at a CAGR of 12 % with 2020 being the only year with negative growth.

Figure 2.4: Visa's revenue 2009 - 2022

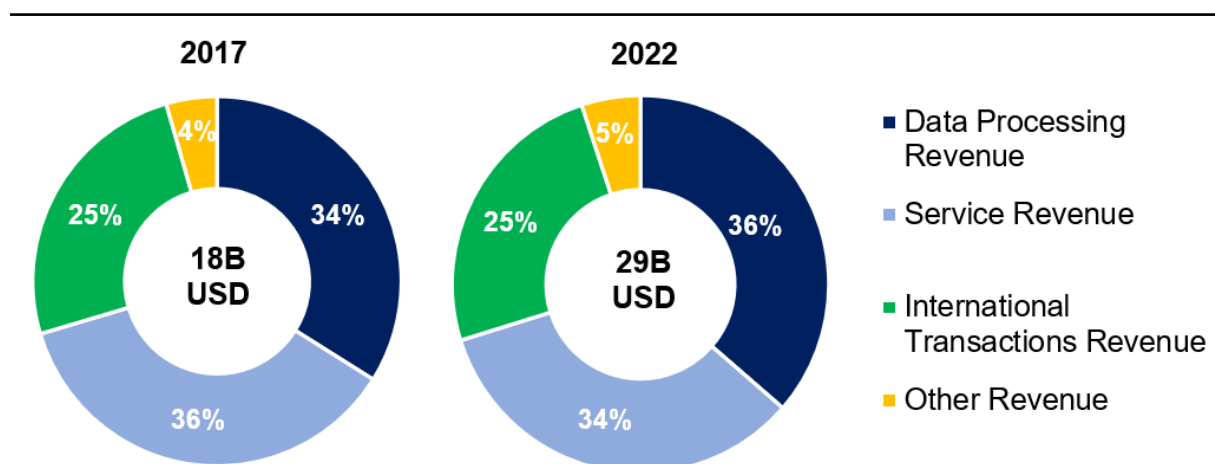


Source: Bloomberg

Business segments

Visa's revenue is split between its three main business segments. In 2022, the biggest at 36 % was Data Processing Revenue which is paid by merchants to Visa for facilitating payments are cleared, settled, and authorized [5]. Service Revenue, almost as large at 34 %, is fees from banks and financial partners paying for support to provide Visa products [8]. Both Visa [5] and analysts [8] consider payment volume in VisaNet to be the primary driver for both the Data Processing and Service Revenue. Finally, International Transaction Revenue made up 25% in 2022 and is paid to Visa when facilitating payments across borders which are extra lucrative as Visa can charge higher fees. In the past five years, the revenue CAGR of the different business segments has been similar at 10 %.

Figure 2.5: Visa's revenue split on business segments: 2017 and 2022

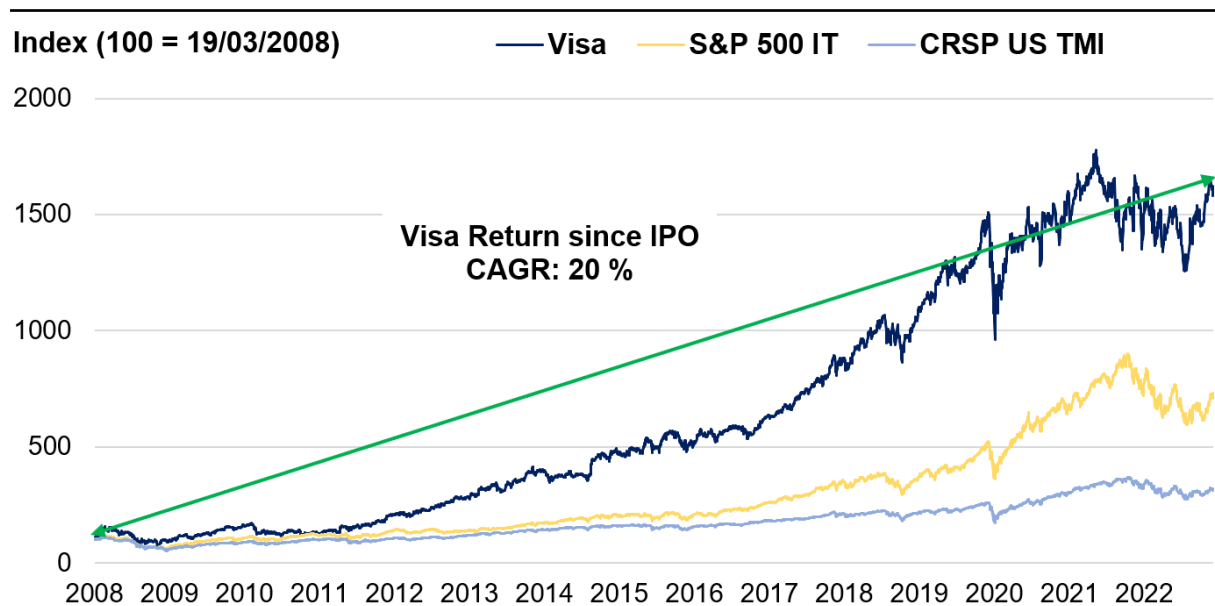


Source: Visa's Annual Reports 2017 [11] & 2022 [5]

2.4 Ownership Structure & Stock Performance

Prior to 2007, Visa was owned by a consortium of banks⁵ which ensured all banks had the incentive to issue payment cards using VisaNet [12]. Visa IPO'd on the 18th of March 2008, the largest US IPO at the time⁶, raising \$17.9B USD [13]. Today, due to Visa's size, the largest shareholders are asset managers, Vanguard (8.6 %), and BlackRock (7.7 %)⁷. Since the IPO, the stock has annualised returned 20 % outperforming the CRSP US Total Market Index (TMI)⁸ which concurrently returned 8 %.

Figure 2.6: Visa's total stock returns since IPO



Notes: 1) Dividends are assumed to be reinvested. 2) Visa is a part of the sub-index S&P 500 IT within the CRSP US TMI.

Source: Bloomberg

To value Visa, it is interesting to look at what major themes according to NASDAQ (2016) [14] and our analysis have impacted the valuation historically:

- 1. 2010 - 2011:** Visa's stock price was turbulent due to uncertainty of how the Durbin Amendment, a part of the Dodd-Frank Act, would affect Visa's revenue as it mandated the Federal Reserve to ensure that debit interchange fees were "reasonable and proportional to the processing costs incurred." In 2011, the Federal Reserve's final ruling was to cap interchange fees at \$0.21 which was received as good news for Visa and resulted in an 11 % stock price increase.
- 2. 2014 - 2015:** Visa attempted to enter the Chinese market, and it initially appeared that China would allow competition with UnionPay, their national payment network. However, as of 2023, Visa (and Mastercard) have not gained any significant foothold in China.

⁵Pre-IPO, Visa was split into four companies, Visa International, Visa US, Visa Canada, and Visa Europe. Each had different ownership structures. Visa US, Canada and International merged in preparation for the IPO and the last twin company, Visa Europe, was acquired through M&A in 2015

⁶In 2014 surpassed by Alibaba

⁷Source: Bloomberg

⁸The CRSP US TMI is a value-weighted index consisting of all US stocks. It is here chosen as the index for consistency purposes as it is the primary market index used when estimating the Cost of Capital. In practice, 80 % of the CRSP US TMI is made up of the S&P500 and the correlation between the two has been 99.5% in the last 10 years

3. **2017 - 2019:** Visa presented strong financial results. The transaction volume in VisaNet doubled from primo 2017 to ultimo 2019 resulting in the revenue increasing 52 %, while both gross and profit margins expanded in the same period. The net income increased by almost 100 % in 3 years.
4. **2020:** Covid-19 significantly decreased private consumption and thereby payment volume in VisaNet. Visa's stock initially decreased by almost 40 % but quickly recovered in line with the rest of the market as large-scale recessions were mostly avoided in Visa's core markets, the US and Europe.
5. **2021 - present:** Visa's stock has been turbulent following increasing inflation and interest rates. In addition to the drop in private consumption from a possible recession, Visa is a growth stock⁹ and sensitive to long-term changes in the discount factor. On the other hand, Visa's business model is naturally an inflation hedge as payment volume is a nominal value.

⁹Defined here as a high market-to-book ratio compared to the CRSP US Total Market Index in line with French [15]

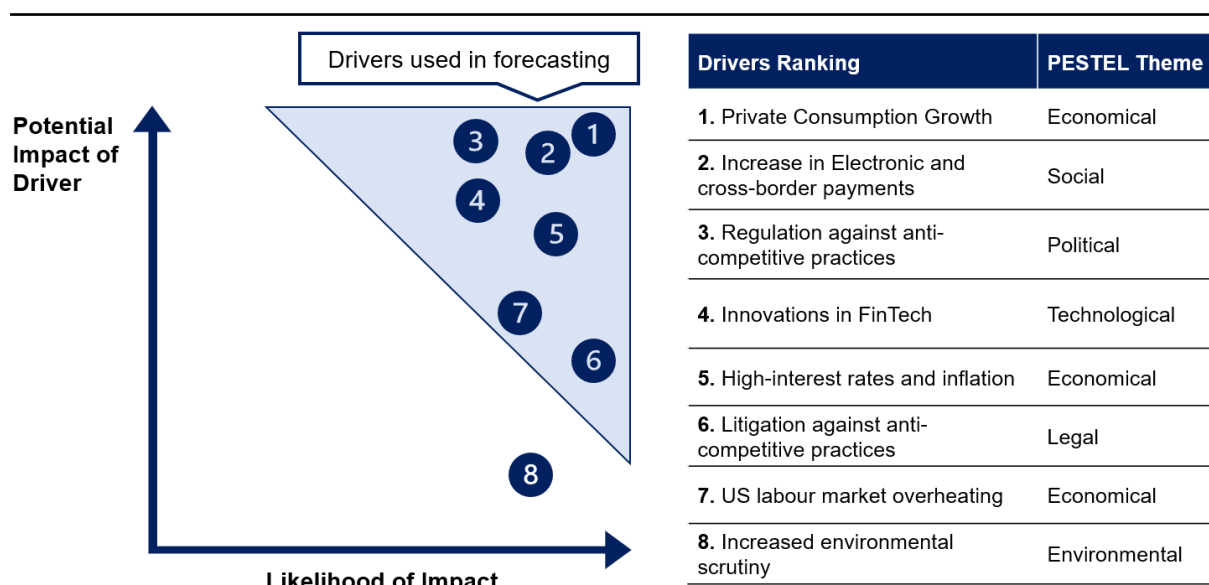
Strategic Analysis

The purpose of the strategic analysis is to enhance the understanding of Visa's strategic position and, combined with the financial statement analysis in Section 4, project well-founded estimates of Visa's financials for the next 10 years in Section 5 [2]. The strategic analysis is based on Visa as a player in the payment network market as described in the company presentation in Section 2. To determine the attractiveness of the payment network market, specifically the market size and projected growth, we analyse relevant macroeconomic drivers in a PESTEL framework. To analyse Visa's competitive situation and project future market shares and profit margins, we use Porter's Five Forces.

3.1 Macro Environment: PESTEL

The PEST framework was developed by F. J. Aguilar (1967) [16] and has more recently been expanded to the PESTEL framework. The framework consists of analysing *political, economic, social, technological, environmental and legal* macroeconomic factors to evaluate the macro-environment a company is operating in. With the PESTEL framework, we have analysed eight drivers with the three most relevant outlined below. Seven of the drivers have been included in the forecasting in chapter 5 and they have all been highlighted in Figure 3.1. Please see Appendix A.2 for the full analysis.

Figure 3.1: PESTEL Drivers: Conclusion on Assessment

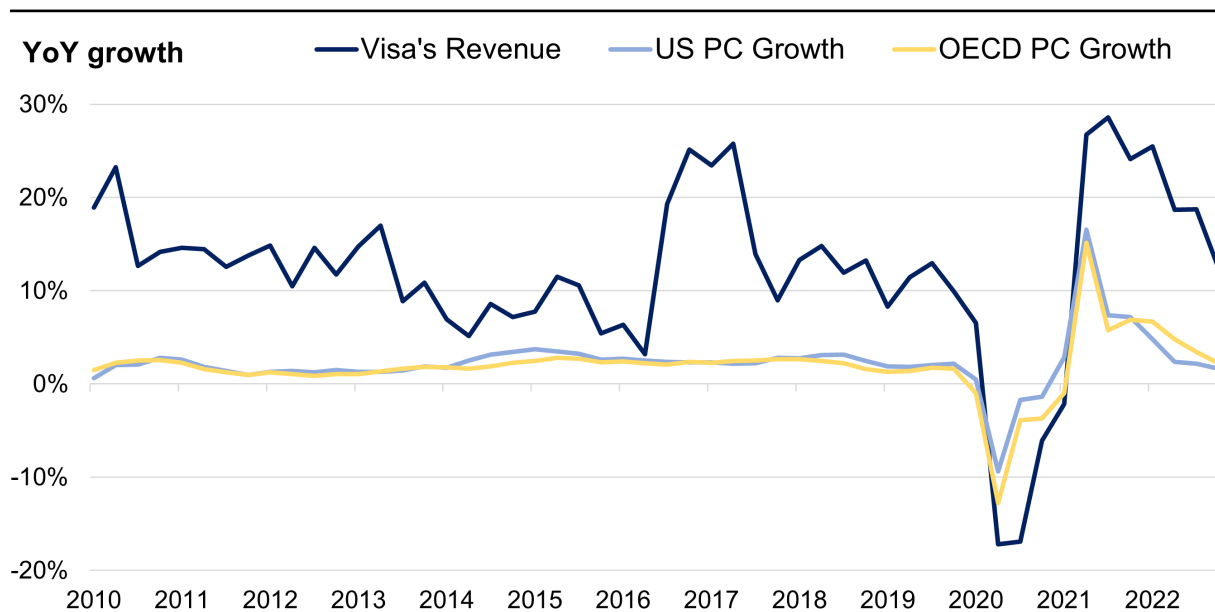


Source: Own Assessment

1. Economical: Private Consumption Growth in OECD and Selected Emerging Markets

As concluded both by Visa in their Economic Outlook Report (2023) [17] and our analysis, the payment network market size is primarily driven by private consumption in especially the US, but also the broader OECD and selected emerging markets [17]. From Figure 3.2, the correlation between Visa's revenue growth and all OECD countries' private consumption has been strong at 0.76 between 2009 to 2022. However, note the revenue growth of Visa has consistently been higher than the private consumption growth in developed markets.

Figure 3.2: Year-on-Year (YoY) quarterly growth in Visa's revenue and OECD private consumption



Notes: Correlation = 0.76 between Visa and OECD PC Growth

Source: Bloomberg, OECD & Own Calculations

While private consumption was severely negatively impacted by the Covid-19 crisis, it has prior been a rather stable¹ macroeconomic indicator. Since the Second World War, the US has only experienced lower than -1% annual private consumption growth which was in 2009 and 2020 [18]. As Visa is a tax on this aggregated stable growth, this is the reason why Visa could have low exposure to cash flow market risk. Our assessment is Visa's revenue will continue to heavily depend on private consumption growth which will return to historically stable growth rates in the next 10 years.

2. Social: Increase in Electronic and Cross-border Payments

The growth of the payment network market is both decided by private consumption and also the share of that private consumption happening electronically. This is why Visa has experienced long-term revenue growth rates above private consumption growth. We assess this can continue in the foreseeable future as the Federal Reserve Payments Study (2022) [19] estimates only 46 % of US private consumption is card payments² and the Bundes Bank (2022) [20] estimates only 42 % of German transactions happen electronically³. In addition, increased online retailing and travel increases cross-border transactions which

¹While there are fluctuations, private consumption has almost been as stable as GDP

²Calculated as total US card payment volume divided by the total private expenditures

³Bundes Bank (2022) states 58 % are cash transactions meaning only 42 % must be electronic transactions

are lucrative for Visa as outlined in Section 2. Overall, we predict the trend toward more digital payments to continue allowing the payment network market to experience significantly higher growth rates than the rest of the economy.

3. Political: Regulation against anti-competitive practices and pricing power

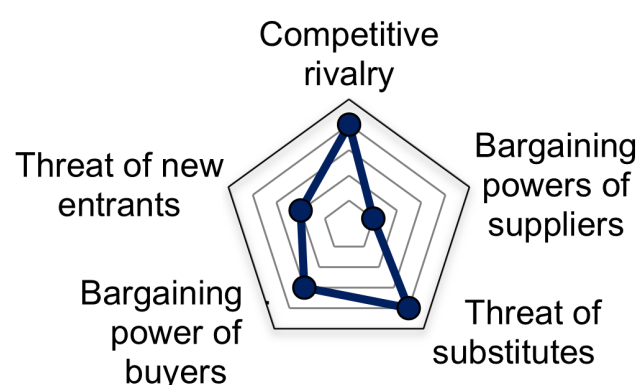
Payment networks, especially Visa and its main competitor MasterCard, have been under political scrutiny by both US and European regulators for anti-competitive practices and too high prices. Analyzing the Credit Card Competition Act of 2022 [21] in the US and the already implemented Regulation on Interchange Fees (IFR) [22] by the European Commission, our assessment is regulation will not force Visa to reduce prices by either a price cap or by forcing additional competition, but Visa is not able to raise prices either with the current political scrutiny.

For emerging markets, Visa's 2022 annual report [5] states "Certain governments, including China, India, Indonesia, Thailand and Vietnam, have taken actions to promote domestic payments systems [...] by imposing regulations that favor domestic providers." We assess China is not a realistic target with recent political tensions with the US, but Brazil, Thailand and India are growth prospects for Visa, especially within cross-border transactions where domestic networks cannot compete.

3.2 Micro Environment: Porter's Five Forces

The Porter's Five Forces framework was first introduced by Porter (1979) [23] and outlines five aspects to consider when analysing the competitive landscape of an industry, here applied to payment networks and specifically Visa. We only go through the main conclusions, but the entire analysis can be found in Appendix A.1. Each of the five forces is ranked on a scale from 1 to 5, with 5 representing the highest competitive threat to Visa. The results are presented in the spider-web Figure 3.3.

Figure 3.3: Conclusion on Porter's Five Forces analysis



Source: Own Assessment

Competitive Rivalry

Rating 4/5: The payment network industry is characterised by two-sided network effects, meaning, for the network to have value it is essential to have both a substantial amount of consumers and merchants.

If the number of consumers with Visa cards increases, more merchants will be inclined to accept Visa cards, (Harvard, 2015 [24]). If more merchants accept Visa cards, the demand of consumers for Visa cards increases. In addition, payment networks have high fixed costs and low variable costs meaning marginal revenue is almost entirely profits⁴.

The presence of a few, but very well-established, competitors, specifically MasterCard, makes the fight for the essential market share intense. Both Visa and Master spend increasing amounts on client incentives [5] to give attractive deals to issuers to use their payment network, (Bloomberg, 2022 [25]).

Furthermore, low-priced national solutions such as "Dankort" in Denmark are difficult to compete with for Visa. However, they are limited by being domestic solutions and are typically combined with an international payment system, such as in Denmark with the "Visa/Dankort". This means VisaNet is still used for lucrative international transactions.

Threat of Substitutes

Rating 4/5: The FinTech and payment solutions space is characterized by high technological advancement and thousands of startups [26] across the world looking to get a part of the high growth rates and profit margins. Visa is actively trying to minimize this risk by ensuring new payment solutions are built on VisaNet (Bloomberg, 2022 [25]). Examples of this include Apple Pay, Google Pay and MobilePay which are not substitutes since all require users to type in their (Visa) card information thus using VisaNet. Crypto, a hot topic in recent years, naturally bypasses payment networks. But here, Visa actively partners with Crypto.com [27], one of the largest cryptocurrency platforms to provide payment cards thereby using VisaNet. Finally, Visa is extremely aggressive in its mergers & acquisition strategy of new companies which also helps ensure they are at the forefront of new potential substitutes.

Bargaining Power of Buyers

Rating 3/5: Visa has a wide range of players that are all to some extent buyers such as merchants, consumers, and financial institutions. The bargaining power of individual merchants and consumers is very low simply due to the two-sided network effects and a lack of competitors. It is difficult to change payment networks if all stores and consumers primarily use Visa and/or MasterCard. However, Visa provides a very homogeneous product. This means large issuing banks do have bargaining power when deciding if they issue Visa cards over MasterCard. In addition, large merchants can to an extent negotiate with Visa such as seen with WallMart [28].

Threat of New Entrants

Rating 2/5: New entrants in the market will be faced with a huge challenge in establishing a payment network that will not diminish due to two-sided network effects. In addition, fixed costs are high to establish a payment network. Another barrier to entry is, a new network needs the brand reputation and know-how to both set up the network and comply with increasingly complex regulation. However, we do consider new entrants a threat if they are backed by governments or other large companies⁵.

⁴Visa's incremental profit margin is estimated to be 80 % [6]

⁵We will again emphasise, they will need to develop a replacement for VisaNet so e.g. Apple Pay, Google Pay and Mobile Pay are not direct competitors because all require the use of card information thus using a payment network such as VisaNet

Bargaining Power of Suppliers

Rating 1/5: Visa does not rely on suppliers to operate VisaNet as it is entirely built and maintained in-house [5]. Visa's suppliers are limited to software for data analytics or internal purposes such as HR systems [29] and they have no bargaining power over Visa.

Financial Statement Analysis

Our strategic analysis shows that Visa operates in a market characterized by a small number of firms with pricing power due to high barriers of entry. This section analyses how being a price setter translates into Visa's financial statements. In addition, Visa's historic accounting key figures are essential in making a reasonable forecast of future cash flows.

4.1 Reformulation of the Financial Statements

As per legislation, Visa's annual reports follow the generally accepted accounting principles (GAAP). This standard set forward by the SEC is not suited for the analysis and thus the financial statements are reformulated to analyst standards as suggested by Penman (2013) [2]. The main objective of the reformulation is to separate the operating activities from the financing activities as value is, as argued by Penman (2013) [2], only created in a firm's operating activities. The detailed process of the reformulation is elaborated in Appendix A.5 together with both the reformulated and reported financial statements.

4.2 Profitability and Growth Analysis

To analyse the profitability of common shareholder equity in Visa, the key figure of interest is the return on common equity (ROCE). This paper, in line with Penman (2013) [2], breaks down the profitability analysis into three levels. Level 1 consists of breaking ROCE into contributions from operational and financing activities:

$$\text{ROCE} = \underbrace{\text{RNOA}}_{\text{Operational Contribution}} + \underbrace{[\text{FLEV} \times (\text{RNOA} - \text{NBC})]}_{\text{Financial Contribution}} \quad (4.1)$$

The definitions of the drivers are given in Table 4.1:

Table 4.1: Definition of Profitability Level 1 drivers

<i>Level 1: Operating Activities</i>	
Return on Net Operating Assets (RNOA):	$\frac{\text{Operating Income}}{\text{Net Operating Assets}}$
<i>Level 1: Financial Activities</i>	
Financial Leverage (FLEV):	$\frac{\text{Net Financial Obligations}}{\text{Net Financial Assets}}$
Net Borrowing Cost (NBC):	$\frac{\text{Net Financial Expenses}}{\text{Net Financial Obligations}}$

Source: Penman (2013) [2]

Level 2 consists of breaking down the operating activities in RNOA further. Level 2 only focuses on the operational activities as per Penman (2013) [2], no value for common shareholders is created from financing activities. RNOA can be broken down into:

$$\text{RNOA} = \frac{\text{Operating Income}}{\text{Net Operating Assets}} = \frac{\text{Profit Margin} \times \text{Revenue}}{\frac{\text{Revenue}}{\text{Asset Turnover}}} = \text{Profit Margin} \times \text{Asset Turnover} \quad (4.2)$$

Lastly, the Level 3 breakdown is of the profit margin into core operating-, other operating-, and unusual income.

Visa's Profitability

The overall ROCE together with level 1-3 breakdowns for Visa is presented in Table 4.2. Generally, the ROCE for Visa is at a very high stable level between 37 to 41 %.

Table 4.2: Profitability and Growth Breakdown

Year	2018	2019	2020	2021	2022
<i>Level 1: ROCE Breakdown</i>					
ROCE	37.49	39.12	40.77	39.00	37.12
RNOA	30.13	31.98	32.76	33.43	34.28
- RNOA Core Operating Income	27.88	31.52	26.54	28.17	37.94
- RNOA Other Operating Income	2.15	2.94	3.01	3.46	4.29
- RNOA Unusual income	0.10	-2.48	3.21	1.81	-7.94
<i>Level 2: OI and NOA Breakdown</i>					
RNOA	30.13	31.98	32.76	33.43	34.28
Profit Margin	53.16	52.15	60.27	57.06	46.04
Asset Turnover	56.68	61.33	54.36	58.59	74.46
1/ATO	1.76	1.63	1.84	1.71	1.34
<i>Level 3: Profit Margin Breakdown</i>					
Profit Margin	53.16	52.15	60.27	57.06	46.04
- Profit Margin Core Operating Income	49.18	51.40	48.81	48.07	50.95
- Profit Margin Other Operating Income	3.79	4.79	5.54	5.90	5.76
- Profit Margin Unusual Income	0.18	-4.04	5.91	3.08	-10.67
<i>Notes: Profit margin is on an after-tax basis</i>					
<i>Source: Visa annual reports 2017-2022. Data in pct.</i>					

Level 1: Return on Net Operating Assets

Return on Net Operating Assets (RNOA) is widely acknowledged as a quality measure of the profitability of a company's operational activities [2]. The contribution from RNOA is for Visa the primary driver for ROCE as financial gearing is very low as outlined in Appendix A.5. Generally, Visa's RNOA has two great qualities in that it is both high and stable in the range of 30-34 % with a geometric mean of 32.49 %. The underlying components do fluctuate in a larger interval. It is clearly the core operating income that contributes the most to RNOA. This is as expected as we, in line with Penman (2013) [2], expect that other operating income will diminish over time. Hence, it is a health check that the primary return comes from the core operating business activities of Visa. It is noticeable that the unusual income RNOA is changing by a large margin each year. The main driver of these changes is exchange rate changes primarily between the US Dollar and the Euro in their subsidiary, Visa Europe.

Level 2 & 3: Profit Margin and Asset Turnover

Breaking down the RNOA to profit margins reveals high and stable profit margins between 46 % and 53 %. While the profit margin was only 46 % in 2022, we can see the core profit margin was 51 %, and the decrease only was a result of the strong US dollar affecting the profit margin on unusual income. The

profit margins are especially high considering that they are calculated on an after-tax basis. As suspected from the Porter Five Forces analysis, being a price setter in a market with high barriers to entry and low marginal costs results in extremely high profit margins.

The final driver, the asset turnover, also increases steadily in the period. This is due to the large economies of scale in the payment network industry which Visa takes advantage of. While the revenue is increasing, Visa does not need to invest significantly in more operating assets. While intangible assets increase, for instance, goodwill from mergers and acquisitions, the incremental user of VisaNet requires very little additional investments.

Forecasting Metrics

The value of a Visa stock from an investor's point of view comes from the free cash flows that Visa can generate in the future. With the reformulation only looking at operational activities, free cash flows (FCF) can be calculated as Operating Income (OI) subtracting changes in Net Operating Assets (NOA) as presented in Equation 4.3:

$$\text{Free Cash Flows} = \text{Operating Income} - \Delta \text{Net Operating Assets} \quad (4.3)$$

This is the formula, we can use when forecasting Visa's free cash flows in Section 5. By rewriting the formulas presented in Table 4.1, the Operating Income and Net Operating Assets can be defined as:

$$\text{Operating Income} = \text{Revenue} \times \text{Profit Margin} \quad (4.4)$$

$$\text{Net Operating Assets} = \frac{\text{Revenue}}{\text{Asset Turnover}} \quad (4.5)$$

This means, when forecasting, we are interested in projecting Revenue, Profit Margins and Asset Turnover. In addition, we wish to remove unusual items and thus only use core earnings in line with Penman (2013) [2] as unusual income such as exchange rates are not within the scope of this paper to forecast. In Table 4.3 the core key figures are presented and these are used in the forecasting in Chapter 5.

Table 4.3: Visa's Revenue, Profit Margin Operating Income and Asset Turnover: 2018 - 2022

Year	2018	2019	2020	2021	2022
Revenue Growth	12.26	11.49	-4.92	10.34	21.59
Profit Margin Operating Income	52.98	56.19	54.35	53.97	56.71
Asset Turnover	56.68	61.33	54.36	58.59	74.46

Notes: 1) The Profit Margin Operating Income in Table 4.3 is the sum of Profit Margin Core Operating Income and Profit Margin Other Operating Income in Table 4.2

2) Profit Margin is on an after-tax basis

Source: Visa annual reports 2017-2022. Data in pct.

Forecasting

In this Section, we forecast Visa’s financial statements for 10 years. Penman (2013) [2] suggests budgeting should be done in the range of 5-10 years as anything less is insufficient and anything longer is too hard to predict. According to Penman (2013) [2], longer budgeting periods are preferred for firms with competitive advantages and high expected growth rates. As both apply for Visa, the forecasting period is chosen to be 10 years.

5.1 Revenue

Due to Visa’s main business model being ”a tax on the transaction volume” in Visanet, the revenue is a product of the average fee and the volume in VisaNet.

$$\text{Revenue} = \underbrace{\text{Average Fee on Transactions}}_{\text{Constant from PESTEL}} \times \text{VisaNet Transaction Volume} \quad (5.1)$$

As argued in the PESTEL analysis, specifically the analysis of political regulation (#3), we project Visa will take a similar average % in assessment fee from VisaNet volume in the next 10 years. This means Visa’s revenue is a function of the Transaction Volume which can again be factored into Visa’s market share and the size of the Payment Network market:

$$\text{VisaNet Transaction Volume} = \underbrace{\text{Visa's Market Shares}}_{\text{Constant from Porters}} \times \underbrace{\text{Private Consumption} \times \text{Share of Card Payments}}_{\text{Payment Network Market Size}} \quad (5.2)$$

From our Porter’s analysis, we project Visa’s US market share to remain constant due to the increasingly fierce competition with MasterCard and substitute products. Because of the increasing number of national payment networks in the EU and South-East Asia, we predict Visa will not be able to capture additional market share in domestic payments but will be the dominant player in international transactions where national solutions cannot compete.

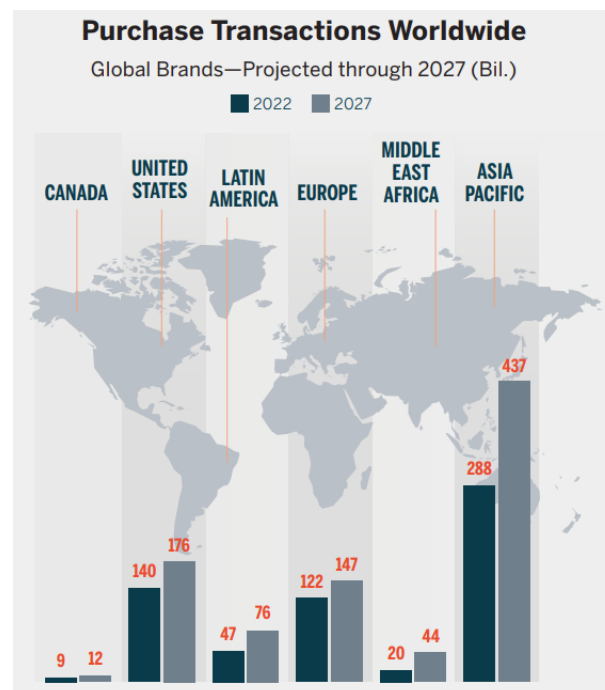
Looking at predicted nominal private consumption growth yields a baseline growth rate in VisaNet volume.

Table 5.1: Nominal Private Consumption Growth: Forecasted 2023-2025

Year	2020	2021	2022	2023F	2024F	2025F
OECD	-2.2	10.2	12.3	7.1	6.9	6.6
India	1.1	19.5	18.1	15.9	16.9	17.2
Brazil	1.1	16.2	10.1	6.5	5.8	5.5

Notes: Realised until 2022. Sum of Private Consumption Volume + OECD CPI Inflation
Source: OECD Economic Outlook 2022-2

On top of this, is the increasing share of electronic payments. As seen in Figure 5.1, leading industry research from Nilsons [9] predicts the number of transactions in payment networks will increase by a CAGR of 4.6 % in the US until 2027, 3.8 % in the EU, and 9.0 % in the Asia-Pacific region.

Figure 5.1: Estimated Growth in Numbers of Transactions in Payment Networks

Source: Nilson Report 1233 [9]

To forecast revenue growth, we have added the contributions from both increased nominal private consumption and the additional number of digital payments. Overall, we find CAGR growth rates of around 10 % in the US and EU and 15% - 25% in the relevant emerging markets. This is fairly well in line with projections by McKinsey (2023) [30] and PwC (2021) [31]. For 2023, we have also incorporated Visa's two released quarterly reports into the projection¹.

5.2 Profit Margin

From the Financial Statement Analysis in Section 4, the operating profit margin has been slightly increasing and reached 56.7 % in 2022. An increasing profit margin has been a characteristic of the Payment

¹However, there is seasonality in quarterly payment volume data so we cannot simply annualize it

Network industry due to the economies of scale and pricing power as argued in the Porter's Five Forces analysis. In addition, technological developments have increased productivity. However, we do not believe the profit margin will continue to increase due to various cost increases on the horizon. First, from Porter's Analysis, competition is intense increasing client incentive costs [5]. Second, Visa will continue to have increased employee costs due to US's red-hot labour market [32]. Third, we believe the litigation expenses will be stable at the current high level. Combining all of these costs, we project a constant profit margin on operations at the current level of 56.7 %.

5.3 Asset Turnover

In the financial statement analysis, Section 4, Visa's asset turnover was shown to be trending upwards with a CAGR of 3.1 % since 2018. This is due to both, from the PESTEL analysis, the market for cashless payments experiencing high growth rates, and also, from the financial statement analysis, Visa does not require capital investments when payment volume increases. However, we do expect Visa's intangible assets to grow, especially the goodwill from acquisitions, as Visa historically has bought small competitors to gain users, increase security in digital payments and win new markets as laid out in the Porter's analysis. This leads us to project an asset turnover growth rate similar to the past 5 years.

5.4 Terminal Growth Rate

The terminal growth rate is very important in the valuation of Visa as a large share of Visa's fundamental value lies here. To better understand the impact on the final valuation of different choices, we wish to explore different scenarios. As described by Penman (2013) [2], the expected nominal GDP growth of the countries where the company is operating is a standard choice. In this case, the company's cash flows are expected to grow at the same rate as the economy. This means if the market is mature and growing by the GDP growth rate, the company is assumed to keep constant market shares.

With this in mind, the base case scenario is a terminal growth in line with the expected nominal GDP growth in the OECD countries and selected emerging markets where Visa has a presence, specifically Brazil and India [18]. The low-growth scenario consists of the expected real GDP growth rate in OECD countries [18] adding only 1.5 % inflation while the high-growth scenario is a case of the real total world GDP growth [18] adding 2 % inflation.

Table 5.2: Terminal growth rates

	Growth Rate
Low growth scenario	3.0 %
Base case scenario	3.8 %
High growth scenario	4.5 %

Notes: All values in (%).

Source: OECD Real GDP long-term forecast and 1.5 - 2 pct. inflation

5.5 Conclusion on Forecasted Drivers

The forecast of the 3 metrics are then:

Table 5.3: Forecast Drivers

Year	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032	T
Revenue Growth	12.1	10.8	9.9	9.4	8.9	8.4	7.9	7.5	7.0	6.5	3.8
Profit Margin OI	56.7	56.7	56.7	56.7	56.7	56.7	56.7	56.7	56.7	56.7	56.7
Asset Turnover	76.7	79.1	81.5	84.0	86.7	89.3	92.1	95.0	97.9	100.9	100.9

Notes: All values in (%). 2023 incorporates 2023Q1 and 2023Q2 report

Source: Own Assessment from strategic- and financial statement analysis

Cost of Capital

The Cost of Capital is the opportunity cost for investors of having money tied up in Visa [33]. If investors did not provide this capital to Visa, they could alternatively allocate their capital to other investment opportunities with similar risk profiles and receive a return [2]. This alternative return is referred to as the **required rate of return** or the **hurdle rate**. The value of future cash flows is subject to discounting at this rate to reflect this opportunity cost [2]. Thereby, the key to determining the cost of capital is *i*) determining what risk factors investors require higher returns for and *ii*) what size premium investors require for taking on these risk factors. As recognized by Morgan Stanley (2023) [33], the risk profile of an investment relates to both the asset class and the company where the capital is placed.

Visa is financed by debt, common equity and preferred equity¹, but the preferred equity share is below 1 % [34]. This means we can significantly simplify the analysis without it impacting the results by converting the preferred equity into equivalent common equity² in line with Visa's equivalence numbers³ [34]. Thus, this paper only looks at debt and common equity financing going forward [33].

6.1 Capital Structure: Weighted Average Cost of Capital

The most common approach to finding the cost of capital of a firm both financed with debt and equity is to use the weighted average cost of capital (WACC) [33]. Here, the cost of equity and debt for the firm are determined separately and then weighed according to the market value of the firm's capital structure. The capital structure of a company is the relationship between debt and equity used to finance assets and operations [35]. Define the market value of debt, D , the market value of equity, E , the cost of debt, r_d , the cost of equity, r_e , and the tax rate, τ , the WACC is given by [33]:

$$WACC = \frac{D}{D + E} r_d (1 - \tau) + \frac{E}{D + E} r_e \quad (6.1)$$

The WACC formula in equation 6.1 reflects that interest paid on debt can be deducted from tax thus making the actual after-tax cost of debt for a company lower than the interest rate paid on its debt. This is known as the tax shield of debt funding [33]. To determine the market cap, E , the required information is the number of outstanding shares and the share price which both are public information for Visa and therefore easily accessible from SEC filings. For the market value of debt, D , Visa has

¹Preferred equity is informally stated a hybrid between equity and debt [33] typically giving additional asset recovery rights than for common equity investors. This means the cost of preferred equity is lower than common equity.

²There is a slight inconsistency as preferred equity was treated as debt in Section 4, but this is simply due to the purpose here being to calculate the cost of capital for the entire Visa enterprise while we only focused on the profitability of common shareholder equity in Section 4

³The preferred equity is a result of Visa's extremely complicated ownership structure due to Visa Europe's pre-merger ownership structure consisting of a consortium of banks. Because of this, Visa offers equivalence numbers for their preferred stock to their freely traded common Class A shares [34]

publicly traded bonds⁴. However, it proves difficult to determine the exact market value of the debt as not all debt is publicly issued bonds and some public bonds have low liquidity. Using the book value of the debt instead is very reasonable as it is set to market value in annual reports⁵ thereby incorporating market-wide changes in interest rates and any potential default risk for Visa into the value of the debt. As shown in table 6.1, Visa's capital structure for the past 6 years has been stable, but tilting slightly more towards equity in recent years.

Table 6.1: Visa's Capital Structure: 2017 - 2023

Year	2017	2018	2019	2020	2021	2022	2023Q2
Debt book value	16,620	16,630	16,690	20,487	21,090	20,920	20,490
Equity market value	266,140	299,350	416,790	482,170	469,420	439,950	487,800
Debt ratio	6%	5%	4%	4%	4%	4%	4%
Equity ratio	94%	95%	96%	96%	96%	96%	96%

Note: Values are ultimo. Values are in Million \$ except equity and debt ratios (in %)

Source: Bloomberg & Visa Annual/Quarterly Reports (2017, 2018, 2019, 2020, 2021, 2022, 2023Q1)

Going forward, we will use the latest capital structure of 96 % equity and 4 % debt for all future years. We see no need to take an average or median of past years as the capital structure has been stable meaning 2023Q2 is an appropriate guess of Visa's future capital structure.

Adjusted Present Value

An alternative to WACC is using adjusted present value (APV) which is typically applied with changing capital structures [33]. The principle is to use the cost of equity as the discount rate for all future cash flows. The enterprise value is then calculated by adding the present value (PV) of the financing effects of debt (mostly the tax shield).

$$\text{APV} = \text{Unlevered firm value} + \text{PV of financing effects} \quad (6.2)$$

Some companies such as Apple and Visa have in the low-interest rate environment issued bonds to fund share buyback programs [36]. A realistic scenario for Visa is they will stop issuing new bonds as their current ones mature due to not finding this approach attractive in the current high-interest rate environment. This would result in a shift towards equity in their capital structure. However, due to Visa's low current share of debt, the impact on the cost of capital and final valuation of incorporating such a scenario is negligible. If the capital structure is assumed constant for all future years, the APV results in the same enterprise value as using an equivalent WACC. Because of this result, we only use the WACC going forward.

⁴We only consider investor-provided debt and not other liabilities in the cost of debt

⁵Financial assets are put at market value in Visa's annual reports [5]

6.2 Cost of Debt

The Cost of Debt (CoD) is the opportunity cost for investors of providing debt financing. Damodaran (2006) [37] suggests two ways of estimating the cost of debt. The first approach is to split the cost of debt into a risk-free rate, r_f , and a spread, r_s , required by investors to take on Visa's credit risk.

$$r_d = r_f + r_s \quad (6.3)$$

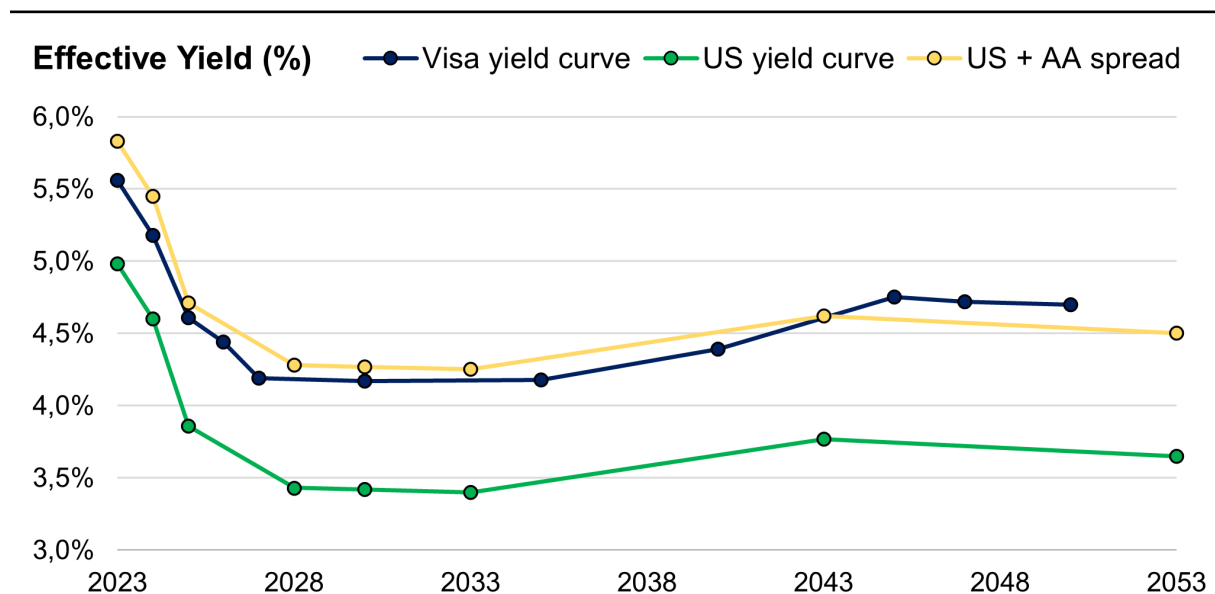
The spread premium is company specific and depends on the firm's credit ratings. Visa's credit ratings are Aa3 (Moody's) and AA- (S&P's) [38]. Damodaran (2023) [39] is generally acknowledged to provide well-researched and realistic spread premiums [33]. Using Damodaran, the AA spread is estimated to be 0.85 %-point.

The second method for estimating the cost of debt is with the effective yield of publicly traded bonds. The effective yield is determined by the coupon rate⁶ and trading prices and is exactly the return investors require to provide Visa debt financing as outlined by Damodaran (2006) [37] and applied by Kumar (2016) [40]. If Visa were to issue new debt tomorrow, the effective yield of their bonds would be the exact price they had to pay for that debt.

To ensure complete comparability between the effective yield of bonds and the cost of debt, it is important to emphasize Visa's public bonds have the same payment structure as US government bonds with no principal payments before maturity and semi-annual interest payments [38]. Visa's bonds vary in maturity, meaning it is possible to create a yield curve for Visa. A majority of the bonds have sufficient liquidity with daily trading, but a few bonds are excluded from the analysis for either having insufficient liquidity or for components making the effective yield not comparable (e.g. different principal payments, being callable).

The most theoretically correct way to measure Visa's cost of debt is thus the effective yield on their publicly traded bonds. As a robustness check, we have also plotted the cost of debt predicted by their credit rating and spread in figure 6.1.

⁶Also the payment structure, but for Visa, this is the same as US bonds

Figure 6.1: Visa's and the US's yield curves

Source: Bloomberg for day 25-04-2023

Visa's yield curve in Figure 6.1 gives an economically very pleasant result since the effective yield on Visa's bonds and the credit spread approach from Damodaran are only different by around 0.1 %-points. When both approaches align, the outcome is compelling as it provides a more reliable approximation of the actual cost of debt. In addition, the bond yields for all maturities are higher than the US yield curve by between 0.4 – 0.7 %-points. This supports that the effective yield of the bonds is a reliable assessment of Visa's cost of debt as investors require a higher return for taking on Visa's default risk compared to US government bonds. Also seen in Figure 6.1, Visa's cost of debt from their bonds is slightly lower than suggested by its credit spread by Damodaran [39] for maturities within the first 20 years. This is also a nice result because there is no clear high liquidity⁷ premium attached to Visa's bonds over US bonds. The spread for Visa's bonds widens slightly with increased maturity which is common for corporate bonds as stated by Damodaran (2006) [37] because default risk increases with longer time horizons.

Moving forward, we use Visa's public bonds as the only approach to estimate the cost of debt with linear interpolation in years without maturing bonds. For the terminal cost of debt, Visa's bond maturing in 2050 is used as its effective yield is representative of Visa's long-maturity bonds. As cost of debt, we only use the effective yield of the bonds, and not the credit spread approach, because *i*) it is theoretically the most accurate number for the return investors require for providing debt financing to Visa *ii*) the effect on the final valuation of trying Damodaran's credit spread approach is very small. This is due to the low share of debt in Visa's capital structure and that the difference between the effective yield on Visa's bonds and the credit spread from Damodaran is only around 0.1 %-points. Finally, the effect on the final valuation of changes in the cost of capital in its entirety is a part of the sensitivity analysis and is thereby tested. Table 6.2 summarises Visa's cost of debt with the current US corporate tax rate [41].

⁷ As stated earlier, all selected bonds have daily trades, but institutional investors might still require a higher return for Visa's bonds as they are not able to sell very large positions in short periods.

Table 6.2: Conclusion on Cost of Debt

Scenario	WACC
Source	Visa's Yield Curve from Figure 6.1
Terminal Value (Visa 20/50 Yield)	4.70 %
US Tax Rate	21 %
Source: Bloomberg for 25-04-2023	

6.3 Cost of Equity Models

The Cost of Equity (CoE) is the alternative cost for investors of providing equity financing to Visa. Any cost of equity model must answer the two questions presented at the beginning of the cost of capital section *i*) what risk factors are investors requiring a higher expected return to take on and *ii*) what is the size of the premium required for this risk. This section explores three different models chosen based on a combination of financial theoretical foundation, frequent use in applied work and, most importantly, relevance for Visa and current market conditions. The main model is the Capital Asset Pricing Model (CAPM). In addition, we introduce the Fama-French factor models from Fama and French (1992) [42] and (2015) [43], and the Good Beta, Bad Beta (GBBB) model by Campbell & Vuolteenaho (2004) [4] based on the intertemporal CAPM (ICAPM).

Capital Asset Pricing Model

The Capital Asset Pricing Model (CAPM) is the most common model for cost of equity used by both academics Cochrane (2005) [44] and practitioners, Morgan-Stanley (2023) [33]. The model was introduced by Sharpe (1964) [45] and Lintner (1965) [46] as the theoretically equilibrium outcome of Markowitz' (1952) [47] mean-variance model. In the CAPM, the expected return of asset i , $\mathbb{E}(r_i)$, consists of the sum of an economy-wide risk-free return, r_f , and the risk premium investors require from asset i , $\beta_i(\mathbb{E}[r_m] - r_{f,t})$.

$$\mathbb{E}[r_i] = \underbrace{r_f}_{\text{Risk-free Rate}} + \underbrace{\beta_i(\mathbb{E}[r_m] - r_f)}_{\text{Risk Premium}} \quad (6.4)$$

The risk premium of an asset i is the product of the asset specific risk exposure β_i and the expected excess return on the overall (stock)⁸ market portfolio. This means in the CAPM, the only relevant risk factor to determine the cost of equity of an asset is β_i which measures how the asset's returns move with market returns [44]. Thereby, the only risk investors require higher expected returns for is market risk [33].

Fama-French Multi Factor Models

A large body of literature has scrutinized the CAPM with the most common empirical criticism being realised stock returns systematically deviate from what is predicted by the CAPM. If long-term realised

⁸Theoretically in the CAPM, the market portfolio contains all asset classes, but is in practice typically limited to public equity due to data constraints

returns can be predicted by any other variable than market risk exposure then the CAPM is not sufficient. As an extension to the CAPM, Fama & French (1992) [42] presented a 3-factor model which, in addition to market risk, includes risk factors Small-minus-Big (SMB) and High-minus-Low (HML). These factors attempt to capture empirically observed higher long-term realised returns for small companies with a high book-to-market value. In Fama-French (2015), they extended the model with the two new factors being Conservative-minus-Aggressive (CMA) and Robust-minus-Weak (RMW) where companies with a conservative investment approach and robust profit margins were shown to have higher returns. Equation 6.5 shows the expected return in the Fama-French 5-factor model:

$$\mathbb{E}[r_i] = r_f + \beta_i(\mathbb{E}[r_m] - r_f) + \beta_{SMB} \cdot \gamma_{SMB} + \beta_{HML} \cdot \gamma_{HML} + \beta_{CMA} \cdot \gamma_{CMA} + \beta_{RMW} \cdot \gamma_{RMW} \quad (6.5)$$

Here, β 's describe Visa's risk exposure to each risk factor and the γ 's are the associated risk premium. We have chosen to apply this model to Visa both because of its frequent use in applied work, but also since Visa as a big, growth company with strong profit margins is exactly the type of company predicted to have a significantly lower cost of equity in the Fama-French models. However, the issue with empirically observed risk factors such as the Fama-French factors, is that there is no clear economic explanation as to why the real cost of equity should be lower for e.g. large firms⁹. This leads to another suggested extension of the CAPM.

Intertemporal CAPM: Good Beta, Bad Beta

Another approach to extending the CAPM with a more robust foundation in financial theory is following the framework of Campbell & Vuolteenaho (2004) [4]. Here, Visa's market risk β_i is split into a cash flow (CF) $\beta_{i,CF}$ and a discount rate (DR) $\beta_{i,DR}$ summing to the CAPM beta:

$$\beta_i = \beta_{i,CF} + \beta_{i,DR} \quad (6.6)$$

The idea of this division in Campbell & Vuolteenaho (2004) [4] is that two elements go into asset pricing/expected returns which are the cash flows of the asset and the applied discount factor. The Intertemporal CAPM (ICAPM) predicts investors require a higher return for taking on cash flow market risk β_{CF} compared to discount rate market risk β_{DR} . The intuition is that an unexpected increase in the discount factor gives lower current asset prices hurting short-term returns, but it also increases the expected return on investments made into future periods. On the other hand, a market-wide decrease in expected future cash flows (e.g. a long recession) only lowers expected returns. This means market risk exposure through discount rate beta β_{DR} is "good beta", as this market risk is priced with a lower required return by investors. On the other hand, cash flow beta β_{CF} is "bad beta" as this part of the market risk is priced with a high required return by investors. Formally, this means expected returns are described by:

$$\mathbb{E}[r_i] = \underbrace{r_f}_{\text{Risk-free Rate}} + \underbrace{\beta_{i,CF} \cdot \gamma_{CF}}_{\text{Cash Flow Risk Premium}} + \underbrace{\beta_{i,DR} \cdot \gamma_{DR}}_{\text{Discount Factor Risk Premium}} \quad (6.7)$$

This is especially interesting for Visa, as they likely have a high discount rate β_{DR} as a growth stock,

⁹Fama & French (2015) does offer explanations such as the factors are proxies for healthier firms, but the multi-factor models have in general been criticised for their lack of foundation in financial theory

especially in recent years with some of the most aggressive interest rate hikes. At the same time, Visa could have a disproportionately lower cash flow β_{CF} due to their business model being a tax on electronic private consumption creating somewhat predictable cash flows as argued in Section 3. The intertemporal aspect is not present in the regular CAPM because it is a static model assuming investors do not make intertemporal decisions regarding savings and consumption because of changes in the discount factor. In CAPM, all market risk is forcibly priced at the same equity risk premium. The two betas are provided in Campbell & Vuolteenaho (2004) [4], but the definitions are changed in this paper to reflect our different implementation method and notation:

$$\beta_{i,CF} \equiv \frac{Cov(r_i, N_{CF})}{Var(r_m)} \quad (6.8)$$

$$\beta_{i,DR} \equiv \frac{Cov(r_i, -N_{DR})}{Var(r_m)} \quad (6.9)$$

Here $N_{CF,t}$ and $N_{DR,t}$ represent news regarding the market cash flows and discount rates. The betas, β_{CF} and β_{DR} , are the covariance between Visa's returns and these news. Note, like Campbell & Vuolteenaho (2004) [4], it is $-N_{DR}$ to reflect discount rates inversely affect stock prices. This definition is significantly simplified from Campbell & Vuolteenaho (2004) [4] in two ways. First, the covariance is here simply between Visa's returns and the news which is a change from Campbell & Vuolteenaho (2004) [4] made to reflect our different estimation approach of $N_{CF,t}$ and $N_{DR,t}$ which is covered in Section 6.4.3. Second, the market variance in Campbell & Vuolteenaho (2004) [4] is only of unpredictable returns the period prior: $Var(r_{m,t} - E_{t-1}[r_{m,t}])$. The simplification made here assumes market returns are unpredictable apart from a constant return, here called μ . Formally this means market returns is governed by $r_{t,m} = \mu + \epsilon_t$ meaning $E[r_m] = \mu$. This is to some extent criticizable, as market returns can somewhat be predicted which is also shown by Damodaran's Implied Equity Risk Premiums in Section 6.4.4. However, due to the limited scope of this paper and some empirical backing from e.g. Cochrane (2005) in market returns generally being difficult to predict, we apply this simplification.

6.4 Estimating Cost of Equity Models

While Section 6.3 presented and motivated cost of equity models, this section implements the models for Visa.

6.4.1 Risk Free Rate

Common to all models is that an economy-wide risk-free rate is required to estimate the cost of equity. As defined by Cochrane (2005) [44], a risk-free asset theoretically requires an investor to know the return of the asset with certainty, meaning the expected return equals the realized in all periods, $\mathbb{E}(r_i) = r_i$. Naturally, no asset is entirely risk-free, but government bonds in developed markets are argued by practitioners, Morgan-Stanley (2023) [33], and the literature, Cochrane (2005) [44], to contain practically no default risk. As a result, the most common choice for the risk-free rate for US listed stocks are US bonds where practitioners favour using the spot price of the US 10 Year (US10Y) bond for all years [33].

However, this is not entirely theoretically correct, as the risk-free rate for a given year precisely should reflect the opportunity cost of not earning the risk-free rate until this specific year, as stated by Damodaran (2023) [48]. Thus, the true opportunity cost is the effective yield on US bonds maturing in the same year the cash flow from Visa is projected. Simply using the US10Y assumes the yield curve is flat at the spot price of the US 10Y. This is not appropriate in this valuation of Visa for a variety of reasons: *i)* As of 25-04-2023, the US yield curve is historically inverted meaning the US 10Y is historically underestimating the risk-free rate for cash flows realised between 2023 - 2033. This means simplifying to a US10Y creates an upward bias in the final valuation of Visa's stock price. *ii)* Using the US yield curve to estimate the forward risk-free rate is simple to implement as there is easily available data on highly liquid US bonds [49]. Therefore, we will use the US yield curve as the risk-free rate as presented in Figure 6.1.

Even when using the US yield curve as the risk-free rate, US bonds yields are no perfect measure of a risk-free asset. Here, we go over the traits assumed of US bonds to be a valid risk-free rate according to Damodaran (2023) [48], and analyse and mitigate potential issues:

No default risk

The risk-free asset should have no default risk. Overall, the US government is considered to have a very low probability of default, but the probability has gained focus with the political turmoil surrounding raising the debt ceiling in the last decade, especially during the spring of 2023¹⁰. In April 2023, the 5-year US credit default swap quotation implies a default probability of 1.06 %¹¹. This is a higher implied default probability than Portugal and Spain. Implied default probabilities are calculated under a risk-neutral probability measure (q) and therefore consist of both the market's assessed real-world (p) probability of default and the market price of that risk which is high according to market participants [50]. Thereby, the market's assessed real-world probability of a US default is likely below 1.06 %. In addition, the market's assessed probability is only high in the short term (1-5 years). This means, while the 5-year CDS spread from Johnson & Johnson (the US company with the highest credit ratings, above the US government) implies a lower default probability of 0.53 %¹², Johnson & Johnson bonds with longer maturity are still trading with a positive spread to US bonds.

Overall, however, the market is clearly not viewing US bonds as default risk-free meaning using US bonds as the risk-free rate estimate might be too high. Damodaran's solution in this scenario is to add a country default risk premium [37]. While Damodaran last suggested premium for the US from primo 2023 was 0 % [51], the market price hike of hedging a US default in the last 6 months has more than doubled. To test the implications of some default risk contributing to a higher effective yield in US government bonds, we will in a low risk-free rate WACC scenario subtract a default risk premium from the risk-free rate¹³. While the CDS price suggested a default risk premium similar to Spain (1.96 %) and Portugal (2.33 %) [51], we have decided to subtract the more conservative default risk premium of the UK (0.7 %) [51] as the UK's economy size and risk premium likely are more comparable with the current US issues than Southern Europe is.

¹⁰The 5-year US credit default swap quotation has increased from 27.5 to 63.4 from January to April 2023

¹¹Source: Bloomberg: the quotation at 25-04-2023 is 63.4 and assuming a standard 40% recovery rate

¹²Source: Bloomberg: the quotation is 25-04-2023 at 31 and assuming a standard 40% recovery rate

¹³The CAPM relies on the existence of a risk-free asset meaning simply changing the rate is a crude way to test implications

No refinancing/interest rate risk

The risk-free asset should have no interest rate risk [48]. Interest rate risk arises if the maturity of the US bonds is not aligned with the cash flows of the stock [48]. In this case, the investor would have to prematurely either buy a new risk-free asset or realize the risk-free asset to the current market price. This creates interest rate risk for US bonds as the market value will deviate from the face value if market interest rates have changed since the bond was purchased. Because of this, it is considered best practice to match the maturity of the bonds to the timing of the cash flows [37], but practitioners often disregard this using the spot price of a US10Y arguing the difference is minimal.

However, with the recent interest rate hikes amplifying this risk and three US banks going into default because of the refinancing risk of US bonds [52], we find it appropriate to eliminate this risk source by matching the maturity of the bonds to the cash flows.

No exchange rate/inflation risk

The risk-free asset should have no inflation or exchange rate risk compared to Visa's cash flows. As US government bonds and Visa's financials are denominated in the same currency, USD, this eliminates any exchange rate or inflation risk.

Conclusion on the risk-free rate

This means we will proceed with the US yield curve, but will keep the US default risk in mind suggesting a slightly lower risk-free rate WACC alternative as well. For the terminal value, we use the yield on the US20Y as we wish to use a yield that reflects maturities beyond 10 years, but also keep in mind most of the terminal value is at the beginning of the terminal period. In addition, the US20Y is a good representative of US effective yields beyond 10 years.

Table 6.3: Conclusion on US Risk-free Rates

Scenario	Main WACC	Low WACC
		Low risk-free rate
Source	US Yield Curve	US Yield Curve – UK's Default Spread
Terminal Value (US20Y)	3.77 %	3.07 %

Source: Bloomberg for day 25-04-2023

6.4.2 Market Risk Beta

Provided by the CAPM, the definition of β_i from equation 6.11 is the covariance between the returns of asset i and the market normalized by the variance of the market returns. The most common approach for estimating β_i is using historical data [33]. This implicitly assumes historical realizations of Visa and market returns are derived from the same distribution as future realizations, specifically that β_i is constant over time [44]. However, a large body of literature has shown β_i is not constant over time [53] which practitioners [44] [33] generally also agree on, but this asks the question of how to estimate β_i into future periods. The most straightforward solution is rolling-window estimations which only use

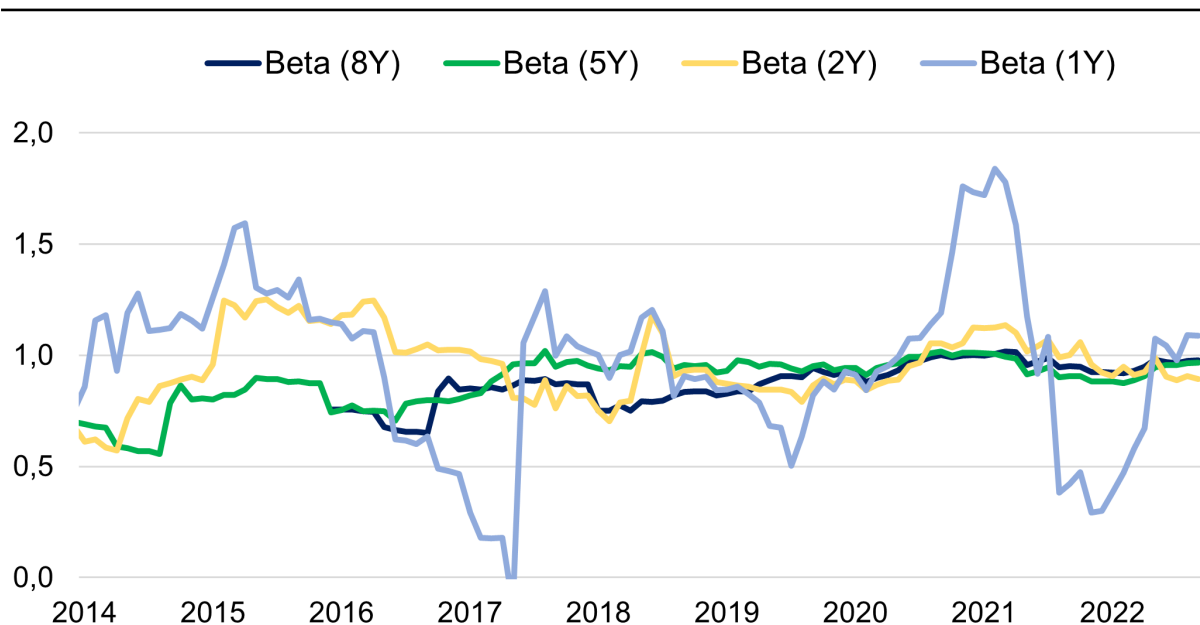
data from a shorter period, e.g. last 5 years, to determine a β_i for each point in time. Formally, β_i can be estimated with OLS or MLE (gives the same result in a linear model with no constraints assuming normally distributed returns) by rewriting the CAPM formula in equation 6.11:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i(r_{m,t} - r_{f,t}) + \varepsilon_t \quad (6.10)$$

Here, α_i is an added constant referred to as Jensen's Alpha [54] or the risk-adjusted excess return as $\alpha_i > 0$ means the return of asset i has been higher than predicted by the CAPM. In practice, using this approach to determine β_i requires selecting the appropriate *i*) market index, *ii*) rolling window period and *iii*) observations frequency.

Damodaran (2023) [55] emphasises, the market index should reflect the investment space the investors of the stock are currently invested into. For Visa, with large institutional owners such as Vanguard and BlackRock, this means a broad world index such as the MSCI World or even the MSCI World All Countries¹⁴. However, to be consistent with the data from the literature our paper builds upon, in particular Campbell & Vuolteenaho (2004) [4] and Fama & French (1988) [56], we use the CRSP US Total Market Index (TMI) which is a value-weighted index consisting of all US stocks, slightly broader than the S&P 500¹⁵. This too is reasonable as Visa is a US stock with primarily US investors highly exposed to the broad overall US public equity market which represents the US economy including real estate through REITS¹⁶. US public equity alone makes up 67 % of the MSCI World Index (2023) [57]. Figure 6.2 displays the evolution of the 8-, 5-, 2- and 1-year rolling beta estimates for Visa with the CRSP US TMI Index and monthly data.

Figure 6.2: Visa's rolling window β_i with CRSP US Total Market Index



Source: Bloomberg on monthly data for day 25-04-2023

¹⁴The MSCI World All Countries includes emerging markets as well

¹⁵In practice, 80 % of the CRSP US TMI is made up of the S&P500 and the correlation between the two has been 99.5 % in the last 10 years.

¹⁶REITS are Real Estate Investment Trusts which are publicly traded investment vehicles invested into real estate

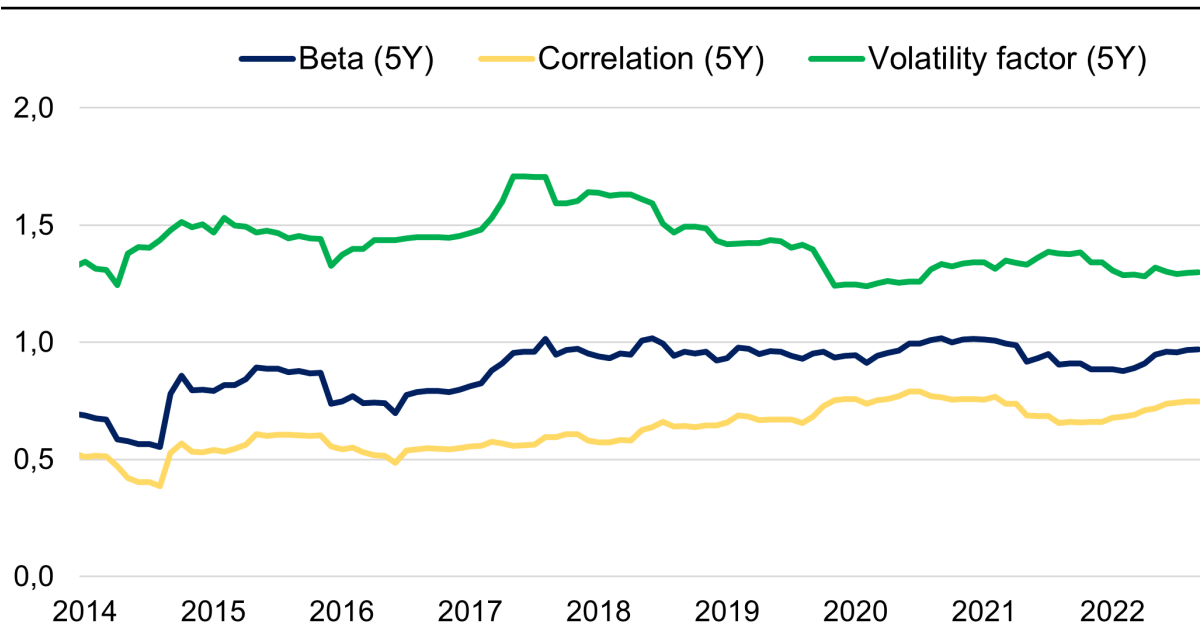
Importantly, the 8-year, 5-year and 2-year rolling window estimates of Visa's β_i have all been stable since 2019 and agree at a beta value just below 1. This means, β_i estimates on the most recent data are not historical outliers which gives an indication it is a good predictor of future β_i values for Visa. Choosing the specific window period is a trade-off between not using old return data which no longer reflects the current β_i of a stock, but also having enough observations to make a reliable estimate¹⁷. Cochrane (2005) [44] states, in practice, the length should depend on the stability of the β_i of the stock. Here, we use the 5-year $\beta_i = 0.97$, but as seen in Figure 6.2, the choice is clearly not of importance. As we are interested in the best guess of a future β_i for Visa, the literature contains suggestions, e.g. Blumes Beta Correction, to adjust the estimated β_i towards 1 to incorporate the mean reverting nature of β_i 's around 1. However, as our estimate is so close to 1, such corrections will have no substantial impact.

The definition of β_i can be separated into *i*) the correlation between the returns of asset i and the market portfolio and *ii*) the volatility of asset i compared to the market portfolio.

$$\beta_i = \frac{Cov(r_i, r_m)}{Var(r_m)} = \frac{\rho_{i,m} \cdot \sigma_i \cdot \sigma_m}{\sigma_m^2} = \underbrace{\rho_{i,m}}_{\text{Correlation}} \cdot \underbrace{\frac{\sigma_i}{\sigma_m}}_{\text{Volatility}} \quad (6.11)$$

Plotting the relationship between the components from equation 6.11 gives Figure 6.3.

Figure 6.3: Visa's 5-year rolling window β_i , correlation and volatility with CRSP US TMI



Source: Bloomberg on monthly data for day 25-04-2023

Both the correlation and volatility have been stable in the last 2-3 years which again confirms our β_i estimate is a good guess of future values. For instance, β_i could have been high due to a temporary period of high returns increasing the volatility, however, there is no sign of such an effect at play here. Going forward, we, therefore, use the 5-year rolling β_i estimation for CAPM of $\hat{\beta}_i = 0.97$ as the main value.

¹⁷The estimator of β_i itself is unbiased for rolling-window estimations, but the reliability of typical standard errors/statistical tests are heavily criticized and disregarded by most academics due to returns empirically displaying ARCH-effects and not being normally distributed

Alternative Data Choices

We have also calculated Visa's β_i on the S&P500 and the MSCI World Index to see the possible consequences of different data choices. In addition, one could also choose different data frequencies. Here, it is a trade-off between having a sufficient number of observations for a reliable estimate, but also avoiding too much noise in the return series from high-frequency data. While we favour monthly data to align with Campbell & Vuolteenaho (2004) [4] and Fama & French (1988) [56], weekly data can be used if the practitioner believes there have recently been structural changes to the β_i value. However, as Visa's β_i has been stable, this is not applied here. Daily data is generally considered too noisy, and Cochrane (2005) [44] even shows it cannot be statistically rejected for all US stocks to have the same expected return on daily data.

Table 6.4: Estimates of Visa's β_i

	Monthly			Weekly		
	2y	5y	8y	2y	5y	8y
CRSP US TMI	0.89	0.97	1.01	0.93	1.02	1.04
S&P 500	0.89	0.91	0.93	0.95	0.98	1.01
MSCI World	0.91	1.14	1.08	0.98	1.05	0.99

Source: Bloomberg

Overall, all estimates are fairly close to 1 in an interval between 0.89 and 1.14. We pick $\hat{\beta}_i = 0.97$ as the main estimate but use $\hat{\beta}_i = 1.14$ from the 5-year MSCI estimation for an MSCI World Index high WACC robustness check.

Using Historic Realizations of the Risk-free Rate to Estimate Beta

From rewriting CAPM in equation 6.11, the historical risk-free rate needs to be subtracted from both the market returns and Visa's returns when estimating β_i . For the historically realized risk-free rate, the appropriate choice for monthly data is 1M US treasury bills¹⁸ as they exactly reflect no default and interest rate risk which is also applied by Cochrane (2005) [44], Fama & French (2015) [15], Campbell & Vuolteenaho (2004) [4]. In practice, for Visa, the impact of subtracting e.g. a US10Y¹⁹ or even no risk-free rate when estimating β_i is negligible²⁰ simply because of the mechanics of subtracting the same number from both the regressor and regressand. However, we align with the relevant literature and use 1M US treasury bills.

Conclusion on Market Risk Beta

Table 6.5 presents the conclusion of estimates of Visa's β_i .

¹⁸Data from Federal Reserve Bank of St. Louis Economic Data is annualized and needs to be converted to monthly yields

¹⁹Bloomberg uses the 10y as the standard country risk premium as many countries do not have data on short-term bonds

²⁰Increased estimated β_i of 0.1 % for Visa

Table 6.5: Conclusion on CAPM beta

Scenario	Main Case WACC	High MSCI World WACC
Index	5-year CRSP US TSM	5-year MSCI World
β_i Estimate	0.97	1.14

Source: Bloomberg

6.4.3 Fama-French and GBBB Betas

The same 5-year rolling window estimation as for the CAPM estimation is applied to Fama-French 3 and 5-factor models. Specifically, this means estimating for e.g. the Fama-French 3 model:

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i(r_{m,t} - r_{f,t}) + \beta_{i,HML}f_{t,HML} + \beta_{i,SMB}f_{t,SMB} + \varepsilon_t \quad (6.12)$$

Here, $f_{t,SMB}$ and $f_{t,HML}$, are the fama-french factors measured as the return difference in period t of portfolios of stocks described by the factor, e.g. the return difference of small minus big companies. The factor data is updated monthly and provided by Kenneth French's website [15]. For the Good Beta, Bad Beta model, the news variables, $N_{CF,t}$ and $N_{DR,t}$, are in Campbell & Vuolteenaho (2004) [4] calculated using a VAR model with logarithms. We have here simplified the method into an OLS time series regression where we first make a regression of Visa's returns, $r_{i,t}$, on the free cash flows to equity (FCFE) predicted for the S&P 500 by Damodaran [55], $FCFE_t$. The point here is to get an estimate of how big the impact on Visa's stock returns is of changes in projected market portfolio cash flows. Specifically, the cash flow news is found by running the regression:

$$r_{i,t} - r_{f,t} = a_{i,t} + N_{CF,t}FCFE_t + u_t \quad (6.13)$$

Please note, using Damodaran's data on the S&P 500 constitutes a consistency issue as we use the CRSP US TMI in the rest of the paper due to it being broader and favoured by Fama & French (2015) and Campbell & Vuolteenaho (2004) [4]. However, the correlation between the two has been 99.5% in the past 10 years²¹ as the S&P500 makes up 80 % of the CRSP US TMI²², and it is not within the scope of this paper to calculate estimated FCFE on another index. This is also an issue later in the equity risk premium estimation, but we simply refer to the explanation here.

The discount rate news, $N_{DR,t}$, can then be found as a residual in the same fashion as in Campbell & Vuolteenaho (2004) by using that unexpected market return shocks must come from either news about the cash flows or discount rates:

$$r_{m,t} - E_{t-1}[r_m] = \hat{N}_{CF,t} - N_{DR,t} \quad (6.14)$$

This calculation method ensures the cash flow and discount rate betas sum to the CAPM beta. As described, we have simplified the model such that $E_{t-1}[r_m] = \mu$ where μ is simply the average market

returns: $\hat{\mu} = \frac{1}{T} \sum_{t=1}^T r_{m,t}$ using a 35-year period to be consistent with later choices. With the news

²¹Source: Bloomberg & Own Calculations

²²Source: Bloomberg

found, the GBBB betas are calculated by the equations presented in Subsection 6.3. Visa's estimated risk exposures from all models are presented in Table 6.6.

Table 6.6: Results: Cost of Equity Models

	CAPM	FF3	FF5	GBBB
$\alpha_i(Constant)$	0.43 (0.44)	0.49 (0.34)	0.51 (0.21)	0.24 (0.15)
β_i	0.97* (0.01)	0.97* (0.01)	0.96* (0.01)	
β_{SMB}		-0.30 (0.23)	-0.25 (0.29)	
β_{HML}		0.05 (0.82)	0.01 (0.92)	
β_{RMW}			0.09 (0.79)	
β_{CMA}			0.04 (0.83)	
β_{CF}				0.16* (0.01)
β_{DR}				0.81* (0.00)
R^2	0.54	0.56	0.56	0.59
T	120	120	120	120
Sample start	2018M04	2018M04	2018M04	2018M04
Sample end	2023M03	2023M03	2023M03	2023M03

Notes: The p-value of a double-sided t-test in (·)

A * signals significance in the double-sided t-test.

As seen in Table 6.6, both Visa's market risk exposure in the CAPM and the cash flow and discount rate betas in the GBBB model are highly statistically significant. The GBBB model has a higher explanatory power of Visa's realised returns measured by R^2 compared to the CAPM. While Visa's low β_{CF} compared to the high β_{DR} appears to confirm our hypothesis that Visa is disproportionately exposed to discount factor market risk, nothing can be inferred here simply due to the method of calculating β_{DR} capturing all residual variation in the unexpected return shocks²³. We can only make such conclusions if the GBBB model actually suggests a lower cost of equity for Visa than the CAPM.

The Fama-French risk factor exposures are, however, not either statistically or economically significant for Visa. Because of this, we will disregard the Fama-French model moving forward. Instead, we use the CAPM as the main case cost of capital while the GBBB is used to explore a scenario where the inherent cost of equity of investing in Visa is actually better described by the intertemporal GBBB model.

6.4.4 Equity Risk Premium

The equity risk premium (ERP), $\mathbb{E}[r_m] - r_f$, is the additional return required by investors to take on the risk of the diversified market portfolio instead of investing in the risk-free asset [2]. As we are interested in discounting Visa's future cash flows, the market risk premium used should be the best guess of the

²³All stocks are expected to be disproportionately exposed to discount factor market risk compared to cash flow market risk with this method. But the interesting part is if Visa is less exposed to cash flow market risk compared to other stocks and if this suggests a lower cost of equity for Visa

ERP in future years [33]. Three methods are suggested by Damodaran (2023) [55] and Morgan-Stanley (2023) [33] to estimate risk premiums, *i*) implied risk premiums, *ii*) historical risk premiums and *iii*) surveys of investors, academics and company executives.

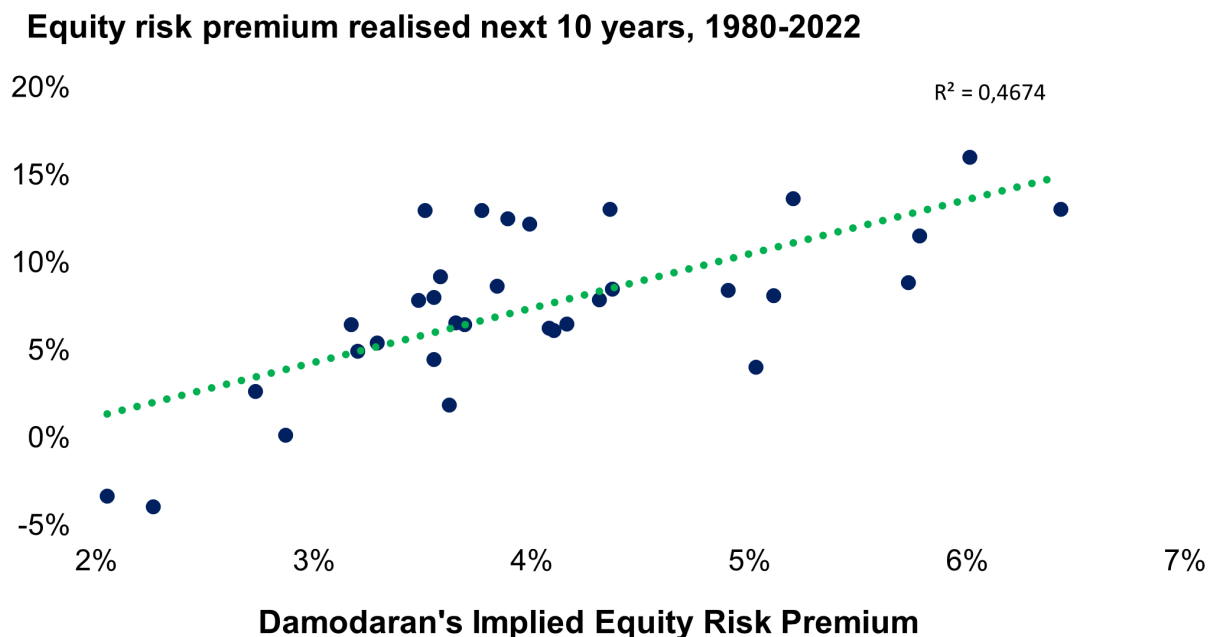
Implied risk premium

By knowing the price and the expected free cash flow to equity of an index, it is possible to calculate what return the investor can expect by purchasing the index as outlined by Damodaran (2023a) [55]. In practice, this means solving the following equation for r :

$$\text{Index Value} = \sum_{t=1}^N \frac{\mathbb{E}[FCFE_t]}{(1+r)^t} + \sum_{t=N+1}^{\infty} \frac{\mathbb{E}[FCFE_{N+1}]}{(r-g)(1+r)^N} \quad (6.15)$$

Naturally, this means estimating all future free cash flows to equity. To do this, Damodaran uses a forecasting period, N , and then adds Gordon's Growth Model for the terminal value afterward with a growth rate g . Using this approach gives an implied ERP. The appealing aspect of this approach is it is a forward-looking estimate of the ERP, a variable emphasised by Damodaran (2023) [55] to not be constant over time. The implied ERP is calculated monthly by Damodaran on the S&P500²⁴. To see if the implied ERP is able to predict the actual realized ERP in the next years, we plot Damodaran's earlier implied ERP against the ERP that was actually realized in the next 10 years and conclude there is a clear positive correlation.

²⁴As described earlier, using the S&P500 poses a consistency issue as we use the CRSP US TMI in the rest of the paper due to it being broader and favoured by Fama & French (2015) and Campbell & Vuolteenaho (2004) [4]. However, the correlation between the two has been 99.5% in the past 10 years as the S&P500 makes up 80 % of the CRSP US TMI, and it is not within the scope of this paper to calculate the implied ERP on another index.

Figure 6.4: Damodaran's implied ERP and realized ERP next 10 years

Notes: 1) The realised ERP is the CRSP US Total Market Index subtracting US1M

2) There is a consistency issue as Damodaran's implied ERP is on the S&P500 while realised ERP is for the CRSP US Total Index, please see Subsection 6.4.3.

Source: Implied ERP from Damodaran and realised ERP data from Kenneth French's website [15]

This predictive power is far superior to both the historical and survey estimation approaches. For this reason, we use Damodaran's implied risk premium in our main WACC. For April 2023, the implied risk premium was 4.88%.

Historical risk premiums

Another approach to estimating the equity risk premium is averaging past realised equity risk premiums. By picking a broad stock index and subtracting the US1M in line with Fama & French (2015) [43] and Campbell & Vuolteenaho (2004) [4], we can calculate the realised equity risk premium. Damodaran (2023a) [55] favours using the geometric mean over long time horizons which is applied here:

Table 6.7: Historic ERP estimates: Geometric Average Total Index Returns above the US1M

	5y	10y	20y	35y
CRSP US TMI	9.8%	10.4%	5.5%	5.4%
S&P 500	10.3%	10.9%	6.6%	6.3%
MSCI World	7.9%	8.2%	6.2%	6.0%

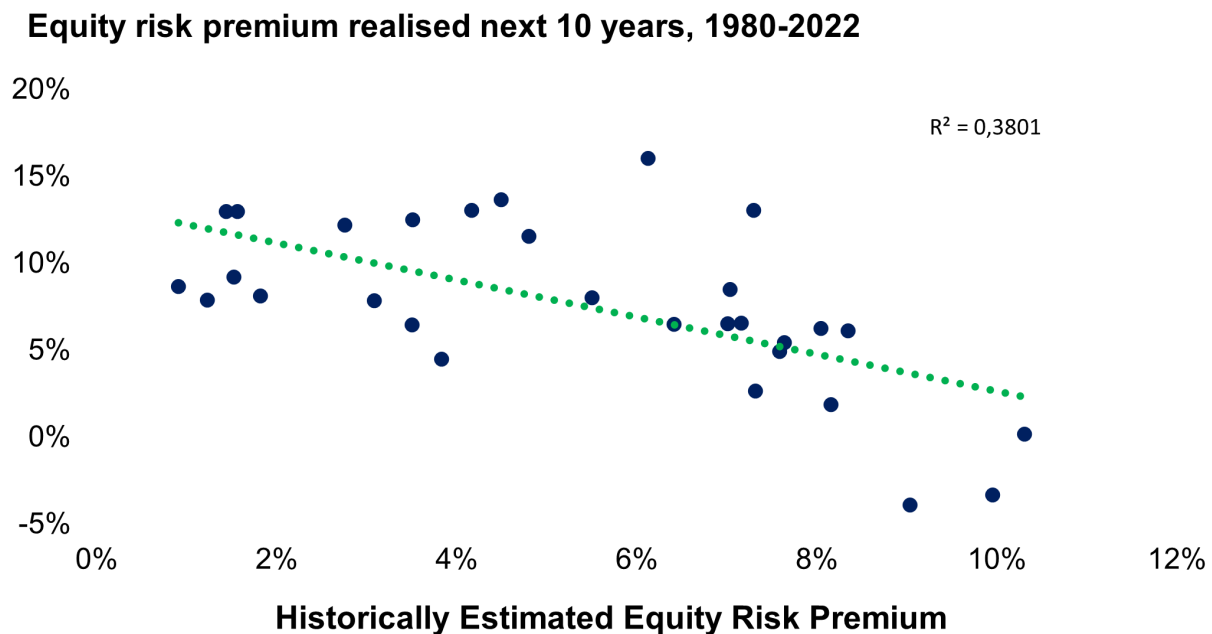
Note: The US1M is subtracted as the risk-free rate.

Source: Bloomberg

Overall, the historic equity risk premiums have been high in the past 10 years. However, the reason why this is not a good estimate of future equity risk premiums is that the predictive power is extremely poor. Plotting the historically estimated ERP from the last 20-years against what it actually turned out to be

in the next 10 years, shows a negative correlation. As stated by Damodaran (2023a) [55], this means only estimated ERP's from realised values over very long periods are appropriate to avoid the negative autocorrelation.

Figure 6.5: Historically estimated ERP from past 20-years and realized ERP next 10 years



Note: There is a consistency issue as Damodaran's implied ERP is on the S&P500 while realised ERP is for the CRSP US Total Index, please see footnote 18.

Source: Implied ERP from Damodaran and realised ERP data from Kenneth French [15]

To test the implications of using the MSCI World as the market portfolio, we use the ERP estimated by the 35-year geometric average of the MSCI World subtracting US1M treasury bills in the MSCI World high WACC scenario. This estimate suggests an ERP of 5.98 %.

Survey risk premiums

As the ERP is the return required by investors to take on market risk, investors or corporate insiders could ideally be asked what their expected equity risk premium is for the foreseeable future. As stated by Damodaran (2023a) [55], the equity risk premium of interest here is that of the marginal investor in Visa, as that investor exactly sets the cost of equity required by the market to invest in Visa. It is naturally impossible to know who the marginal investor is, but Damodaran (2023a) [55] suggests using the largest shareholders. BlackRock (2023) [58] states their return expectations of US stocks in the next 20 years are 7.5 % while their expectations for US government bonds are 3.1 %. This means an ERP of 4.4 %. However, ERP's estimated by survey are stated by Damodaran (2023a) [55] to have low or even negative predictive power and generally be unreliable by, for instance, being overly sensitive to recently realised returns. Therefore, we will not use the survey risk premiums. However, note the stated ERP from BlackRock is in line with the implied ERP of 4.88 %, although slightly lower which would result in a higher valuation.

Conclusion equity risk premium

The main WACC scenario will use the Implied Equity Risk Premium while a high WACC scenario will use the MSCI World 35-year estimate.

Table 6.8: Conclusion on ERP

Scenario	Main Case WACC	High MSCI World WACC
Source	Implied Equity Risk Premium	MSCI World (35Y)
ERP Value	4.88 %	5.98 %

Source: Damodaran Implied ERP for April 2023

6.4.5 GBBB Risk Premium

Due to our different estimation method in the GBBB betas and the results of Campbell & Vuolteenaho (2004) [4] being outdated, we have estimated the risk premiums attached by the market on cash flow and discount factor market risk with updated data. This is done in a similar fashion as in Campbell & Vuolteenaho (2004) [4] by using the 25 size and book-to-market portfolios from French's website [15], calculating $\beta_{CF,i}$ and $\beta_{DR,i}$ for all portfolios. Using Fama-Macbeth (1973) [59] regressions, also known as the two-pass estimation method, risk premiums are determined by running cross-asset regressions and taking the average of the estimated risk premiums over time. Specifically, after the betas are estimated, 25 regressions (one for each portfolio in the French data [15]) are run:

$$r_{i,t} - r_f = \alpha_i + \hat{\beta}_{i,t,CF}\gamma_{t,CF} + \hat{\beta}_{i,t,DR}\gamma_{t,DR} \quad (6.16)$$

Then the final estimate of the risk premiums is a simple average over time, e.g. $\frac{1}{T}\sum_{t=1}^T \gamma_{t,CF}$. Again, we choose a time horizon of 35 years.

Table 6.9: Annual GBBB Risk Premiums: 35 Years

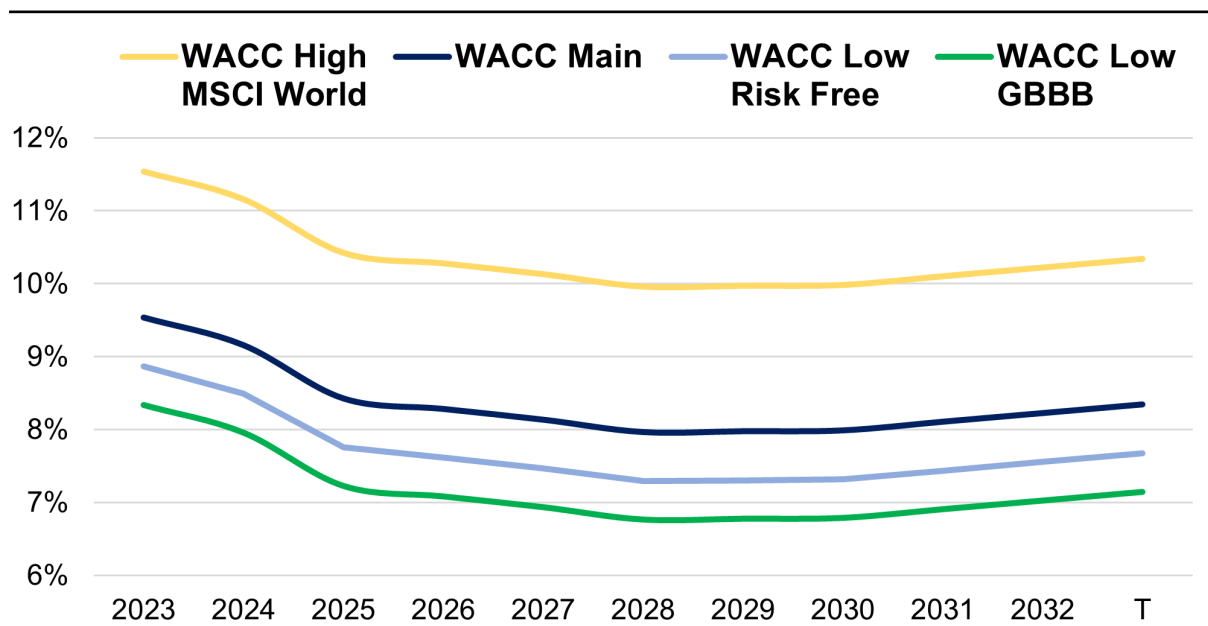
Risk premium	$\hat{\gamma}_{CF}$	$\hat{\gamma}_{DR}$
Annual size	12.02 %	1.92 %

Source: Own Calculations on French data [15]

As theorized, the risk premium attached to the cash flow market risk is significantly larger than the discount rate risk. However, our $\hat{\gamma}_{CF}$ is very low compared to the 36 % estimated in Campbell & Vuolteenaho (2004) [4], but the two are impossible to directly compare due to the different approaches.

6.5 Conclusion on Cost of Capital

Concluding on Visa's cost of capital, the different scenarios are plotted across the budgeting and terminal period. Each scenario represents a vertical shift from the main WACC derived through the straightforward CAPM approach. All scenarios are 'u' shaped which is a result of the US yield curve currently being inverted until around 2029.

Figure 6.6: Visa's WACC on high, main and low cases

Source: Bloomberg & Own Calculations

As predicted, the GBBB alternative to CAPM provides a significantly lower cost of capital for Visa. On the other hand, the high WACC MSCI World is significantly higher than all other scenarios especially because of the higher ERP, but also the slightly higher β_i . All components used in the cost of capitals are for convenience summarised in Table 6.10.

Table 6.10: Terminal Cost of Capital

WACC Scenario	High WACC	Main WACC	Low WACC	GBBB WACC
Derivation of WACC	MSCI World	CRSP US TSM	CRSP US TSM	CRSP US TSM
<i>Cost of Debt</i>				
Effective Yield	4.7 %	4.7 %	4.7 %	4.7 %
Tax rate	21 %	21 %	21 %	21 %
After Tax Cost of Debt	3.7 %	3.7 %	3.7 %	3.7 %
<i>Cost of Equity</i>				
Risk-free rate	3.77 %	3.77 %	3.07 %	3.77 %
β_i	1.14	0.97	0.97	
β_{CF}				0.16
β_{DR}				0.81
Equity Risk Premium Market	5.98 %	4.88 %	4.88 %	
Equity Risk Premium CF				12.02 %
Equity Risk Premium DR				1.92 %
Visa's Risk Premium	6.82 %	4.73 %	4.73 %	3.48 %
Terminal Cost of Equity	10.59 %	8.50 %	7.80 %	7.25 %
Capital Structure				
Debt Ratio	4 %	4 %	4 %	4 %
Equity Ratio	96 %	96 %	96 %	96 %
Terminal WACC	10.34 %	8.34 %	7.67 %	7.14 %

Source: Bloomberg & Own Calculations

Valuation

7.1 Discounted Cash Flow Valuation

We calculate Visa's common equity value using a discounted cash flow (DCF) model. Using the projected revenue growth, profit margin, and asset turnover from Section 5, allows us to calculate the free cash flows using equation 4.3. The free cash flows in the budgeting period are discounted using the WACC from chapter 6 while the value of the terminal growth is calculated by Gordon's growth model, $V_t = \frac{FCF_{\text{terminal year}}}{WACC - g}$, where g is the terminal growth rate in FCF's. This results in Visa's enterprise value from which net debts are subtracted to find the common equity value.

Table 7.1 shows how Visa's common shareholder value is calculated using the main WACC and the medium terminal growth rate from Section 5.

Table 7.1: Valuation using discounted cash flow model, Medium terminal growth and main case WACC

(USD Million)	2023	2024	...	2031	2023	Terminal
Free Cash Flow	11,897	14,635	...	28,837	31,056	32,025
Discount Factor	0.92	0.83	...	0.50	0.46	
Present value FCF	134,517					
Present value Terminal year	318,858					
Enterprise value	453,375					
Net Financial Obligations (Net Debt)	2,667					
Equity Value Common shareholders	450,708					
Number of Outstanding Shares	2,090,000,000					
Share Price, USD	215.65					

Notes: Calculations are done with a medium growth scenario and main scenario cost of capital. Data in USD million unless stated otherwise. 70 pct. of Enterprise Value is from terminal growth
Source: Own calculations and Visa Financial Statement 2017-2022

The result in Table 7.1 is a share price of 216 USD. Compared to the closing price on April 25th 2023 of 230 USD, it is 6 % lower. This is close considering the many uncertainties of the valuation. A similar calculation using the WACC from the Good Beta Bad Beta model results in a share price of 299 USD.

Believing that GBBB is the correct way of calculating the required rate of return thus means Visa is undervalued as opposed to CAPM which says that the market slightly overvalues Visa.

In order to ensure the DCF model is correctly applied for the budgeting and terminal period, we also calculated the equity value using the residual income model. Theoretically, these two models should give the same value if correctly performed, Penman (2013)[2]. The result from the Residual income model is presented in appendix A.12 and the valuation is equal to that of the DCF model.

7.2 Sensitivity Analysis

The terminal growth rate and the cost of capital are essential in the valuation, especially for a growth stock such as Visa with high cash flows long into the future. Using the three terminal growth rates from Section 5 and the WACC scenarios from Section 6, the share prices are calculated in Table 7.2.

Table 7.2: Sensitivity Analysis using different WACC and growth scenarios

WACC	High	Main	Low	GBBB
Low: $g = 3.02$	136.26 (-40.65 pct.)	194.25 (-15.39 pct.)	225.10 (-1.95 pct.)	256.70 (11.81 pct.)
Medium: $g = 3.80$	145.09 (-36.80 pct.)	215.65 (-6.07 pct.)	255.87 (11.45 pct.)	299.19 (30.32 pct.)
High: $g = 4.50$	155.03 (-32.48 pct.)	242.25 (5.51 pct.)	296.36 (29.08 pct.)	358.68 (56.23 pct.)

Notes: Underneath is the deviation in pct. from closing price on 25/04/2023, USD 229.59

The direction of the stock value changes is intuitive, a higher discount rate and lower terminal growth rate reduce the estimated intrinsic value and vice versa. Worth noticing here, is how sensitive the estimated value is to reasonable changes in the growth rate and the WACC. Simply using the broader, and perhaps more appropriate, MSCI World Index to estimate beta and risk premiums in CAPM, the fundamental value is only 145 USD, 37 % below the trading price on 25-04-2023. Visa's share value does not react as sharply to the different terminal growth values found reasonable to apply in this paper, but we can see it is still very important to the valuation. Noticeably, the GBBB WACC results in Visa being undervalued by the market for all terminal growth cases.

7.3 Relative Valuation: Multiple analysis

As a robustness check on the DCF valuation, we wish to measure Visa's common shareholder value using relative valuation. This is, according to Yee (2004) [60], desirable as no single valuation model gives the correct answer. To conduct the valuation, we use both core and superficial peers as introduced in Section 2 as relative valuations require a large sample of peers to avoid outlier results influencing the valuation [2]. However, it should be kept in mind that when conducting relative valuations, peers should have the same cost of capital, profit margins, and growth outlooks [2]. This is not fulfilled here as Gateways, Acquirers and smaller payment networks are not able to compete with Visa's profit margins and

potential growth due to the high barriers to entry and economies of scale in the payment network industry.

The chosen multiples are the Forward Price-Earnings (P/E), Enterprise Value/EBITDA (EV/EBITDA), and the Price-Sales (P/S). The forward P/E multiple is the most commonly applied [2] and measures how much 1 dollar of next year's profits in the company costs [2]. Because of this, high growth companies typically have high (P/E) ratios as their earnings are expected to grow in the future. The same expectation is present if the EV/EBITDA multiple is high. EV/EBITDA is independent of capital structure as it measures the entire enterprise value against cash flows where no creditors or investors have been paid yet. These two multiples are both accrual-based which according to Liu (2002) [61] is superior compared to cash-flow-based multiples. Further, we use forward multiples as they incorporate expected growth in the next year thereby using all available information. The multiples are presented in Table 7.3.

Table 7.3: Multiples of Peers

Peer	Forward Price/Earnings	EV/EBITDA	Price/Sales
Visa	28.11	21.27	15.58
MasterCard (Payment Network)	32.33	24.38	16.32
American Express (Payment Network)	13.75		2.18
Discover Financial Services (Payment Network)	7.40		2.46
Fiserv (Acquirer)	25.74	12.11	4.20
Adyen (Acquirer)	69.18	43.48	4.96
Paypal (Gateway)	18.20	9.18	2.51
Block/Square (Gateway)		24.91	1.85
Harmonic Average	17.29	16.91	3.04
Visa Market Cap (Multiple calculated), USD Mil.	275,000	372,978	93,002
Visa Share Price, USD	131.58	177.18	44.50
Median	21.97	24.38	2.51
Visa Enterprise Valuation, USD Mil.	349,459	537,717	76,872
Visa Share Price, USD	167.21	256.00	36.78

Source: Trading multiples are found on Seeking Alpha at 25/04-2023

Visa is valued using a harmonic average and a median to reduce outlier bias and overvaluations as stated by Baker (1999) [62]. From Table 7.3, the valuation of Visa is low with the forward P/E and incredibly low using P/S. However, the EV/EBITDA multiple produces a valuation larger than our main case DCF. Overall, we do not put emphasis on the results as the spread in the multiples clearly reflects Visa's lack of close peers. This can especially be seen in the Price-Sales multiple where Visa's and MasterCard's high profit margins ensure they are in a league of their own¹. Compared to MasterCard, Visa's only true peer, Visa's shares are slightly cheaper measured by all three multiples.

¹This is also the reason more elaborate peers analysis, e.g. finding an industry beta, is not applied in this paper and should only be applied to Visa with caution

Conclusion

To find the fundamental value of common shares in Visa, this paper projects Visa's future cash flows based on both a strategic and financial statement analysis. The strategic analysis concludes, Visa has competitive advantages in the very attractive payment network market which is characterised by high growth rates and few companies with pricing power. This is translated into the financial statement analysis, which shows Visa to be extremely profitable with a return on common equity at 37 % and profit margins exceeding 50 %.

Visa's terminal cost of equity is estimated to be 8.5 % using the regular CAPM resulting in a valuation of 216 USD per share. At the same time, the GBBB model estimates a cost of equity of 7.3 % giving a much higher valuation of 299 USD. Both use Visa's bonds as the cost of debt. The share price of 230 USD as of 25-04-2023 is more in line with the valuation predicted by the CAPM. Assuming the market price is trading at Visa's intrinsic value, this suggests CAPM is the appropriate model for Visa. On the other hand, Visa's realised returns are better explained by the GBBB model measured by the R^2 in the regression. This suggests that GBBB might be a good alternative model to apply to Visa and other stocks sharing similar characteristics of being disproportionately highly exposed to discount factor risk. A multiples valuation shows Visa to be slightly cheaper than MasterCard, but Visa's lack of close peers when sorting on both business model and size makes relative valuations difficult.

Finally, the results of this paper are shown to be incredibly sensitive to different estimation methods and data choices in both the terminal growth rate and the cost of capital. For instance, using the MSCI World Index to estimate beta and the ERP instead of the CRSP US TMI results in a terminal WACC of 10.3 % giving a stock price of only 145 USD, 30 % below the current trading price. Because of this, conclusions on the pros and cons of applying the different cost of equity models to Visa can only be used with the high uncertainty of the results in mind.

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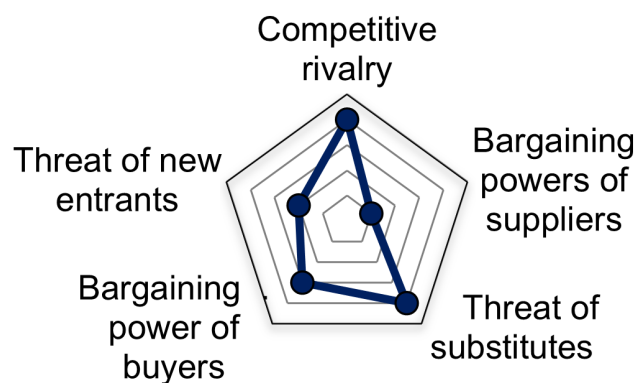
Appendices

Appendices

A.1 Micro Environment: Porter's Five Forces

The Porter's Five Forces framework was first introduced by Michael E. Porter (1979) [23]. The framework outlines five important aspects when analysing the microenvironment and competitive landscape of Visa's business activities. Each of the five forces is ranked on a scale from 1 to 5 and the result of the assessment is presented in the spider-web Figure A.1.

Figure A.1: Conclusion on Porter's Five Forces analysis



Source: Own Assessment

Intensity of Competitive Rivalry

Rating 4/5: The payment network industry is characterised by two-sided network effects, meaning, for the network to have value it is essential to have both consumers and merchants using the network. If the number of consumers with Visa cards increases, more businesses will be inclined to accept Visa cards. If more merchants accept Visa cards, the demand of consumers for Visa cards is higher. In addition, payment networks have high fixed costs and low variable costs meaning marginal revenue is almost entirely profits¹. This means market shares are everything when competing to win in the payment network market. As all of Visa's main peers have well-established networks their focus is to utilize the two-sided network effect and get issuers to provide payment cards to consumers.

The presence of a few, but very well-established, competitors, specifically MasterCard, makes the expenses to client incentives increasing. The expenses stem from increasingly having to spend money on giving attractive deals to issuers to use their payment network.

¹Visa's incremental profit margin is estimated to be 80 % [6]

Furthermore, many countries favour low-priced national solutions for domestic payments, for example, "Dankort" in Denmark and "Girocard" in Germany which are difficult to compete with for Visa. However, they are limited by being domestic solutions and are typically combined with an international payment system, such as in Denmark with the "Visa/Dankort". This means VisaNet is still used for lucrative international transactions.

Bargaining Powers of Suppliers

Rating 1/5: The payment network services that Visa offers do not rely on any suppliers as all the tech services required to run the network are built and maintained in-house. Visa's suppliers are limited to software systems for internal purposes such as data analytics and HR-systems [29]. The market for software solutions is characterized by large competition reducing the price and bargaining power for the suppliers. Furthermore, Visa, as one of the largest US companies, both have a good relationship and large bargaining position with its current suppliers, [6]. While its highly unlikely to be needed, Visa even has the expertise and capital to develop in-house IT systems.

Threat of Substitute Products

Rating 4/5: The FinTech and payment solutions space is characterized by high technological advancement and thousands of startups across the world. This is likely because the payment solutions market has an attractive combination of high growth and profit margins which will be highlighted in section 4. While this naturally constitutes a high rating, Visa is extremely effective at minimizing this risk. First, It is important to emphasise, Visa will only be substituted if a new product outside the current payment network replaces Visa. As stated Visa actively minimizes this risk by partnering up with startups and other companies in the payment ecosystem to make sure new solutions are built on VisaNet. Examples of this is include but are not limited to Apple Pay, Google Pay and MobilePay which are not substitutes since all require users to type in (Visa) card information thus using VisaNet. Another example is crypto which naturally bypasses payment networks. But here, Visa partners with Crypto.com, one of the largest cryptocurrency platforms in existence. Finally, Visa is extremely aggressive in its mergers & acquisition strategy of new companies which also helps ensure they are at the forefront of new potential substitutes.

Bargaining powers of buyers

Rating 3/5: Visa provides its payment network to vastly different buyers such as merchants, consumers, and financial institutions. The bargaining power of individual merchants and consumers is generally not high simply due to the lack of competitors. It is difficult to change payment networks if all stores and consumers primarily use Visa and/or MasterCard. However, large issuers in the form of financial institutions do have bargaining power when deciding if they should issue Visa cards over MasterCard. In addition, large merchants such as WallMart can to a much greater extent set up collaboration between merchants and negotiate with Visa and MasterCard. Issuing banks and large merchants have bargaining power as a result of Visa providing a very homogeneous product, meaning few merchants, consumers or financial institutions would mind switching to MasterCard or other payment networks if it is profitable. As large issuers and merchants have negotiated price discounts with Visa [29], we do consider the bargaining power of buyers a significant factor in decreasing the attractiveness of the industry.

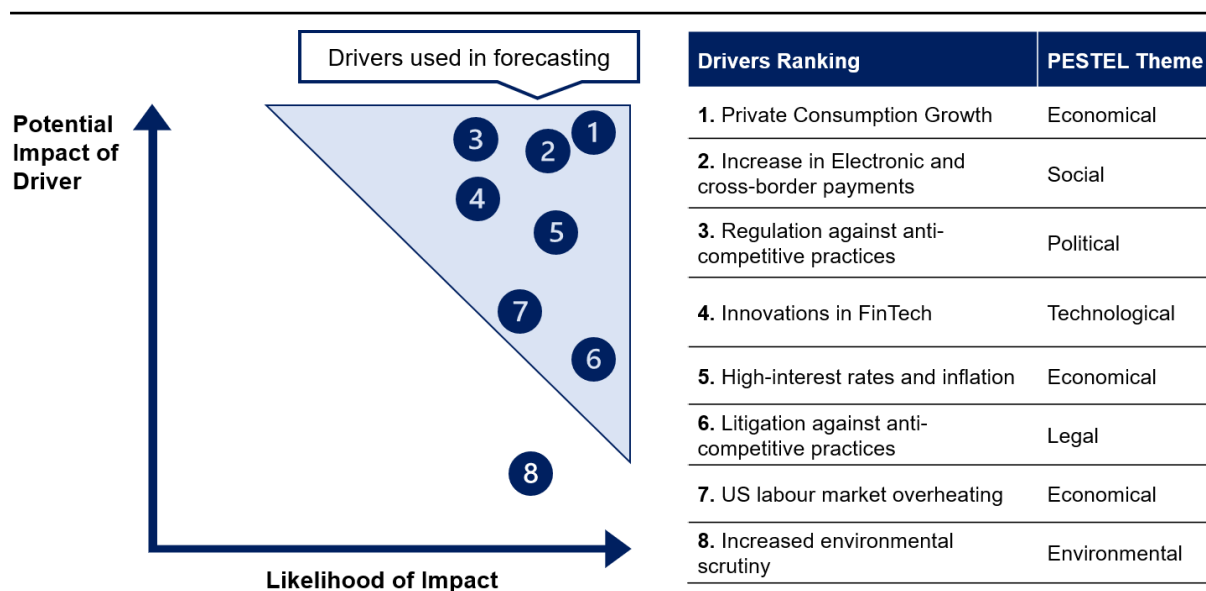
Threat of New Entrants

Rating 2/5: New entrants in the market will be faced with a huge challenge in establishing a payment network that can gain the necessary market share to compete in a market with such high double-sided network effects. For a new potential network to gain a sufficient amount of consumers and merchants, they will likely be running with losses because of the high fixed and low variable costs characterising the industry. This means there are large capital requirements for new entrants. In addition, a new network will need a brand reputation and the know-how to both set up the network and comply with increasingly complex regulations. However, we do consider new entrants a threat if governments or other large players develop their own payment networks².

A.2 PESTEL

The PEST framework was developed by F. J. Aguilar (1967) [16] and has more recently been expanded to the PESTEL framework. The framework consists of analysing *political, economical, social, technological, environmental and legal* macroeconomic factors to evaluate the macro-environment a company is operating in. In this paper, the PESTEL analysis is used to forecast the size, growth and profit margins of the Payment Network market. We have analysed 8 drivers, but have only included the conclusion of the 4 most relevant used in the forecasting of financials as presented in figure A.2. Please see Appendix A.2 for the full analysis.

Figure A.2: PESTEL Drivers: Conclusion on Assessment



Source: Own Assessment

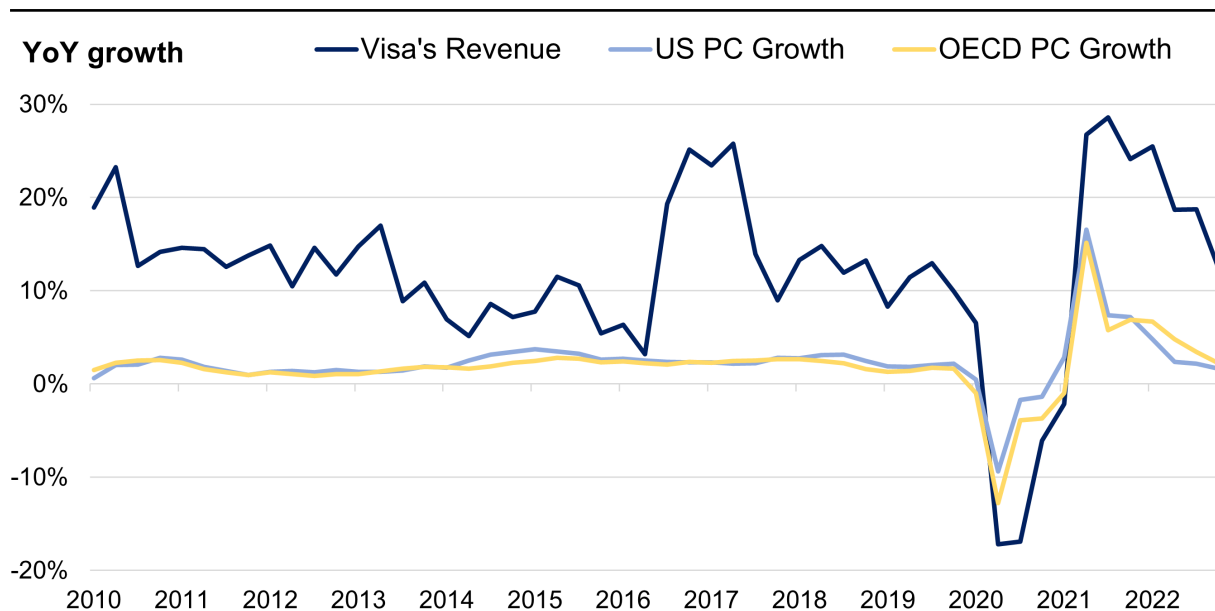
1. Private Consumption Growth in OECD and Selected Emerging Markets

As both concluded by us and Visa in their Economic Outlook (2023) report, the payment network market size is primarily driven by private consumption in especially the US, but also the broader OCED and

²We will again emphasise, they will need to develop a replacement for Visa so e.g. Apple Pay, Google Pay and Mobile Pay are not direct competitors because all require the use of card information this using a payment network such as VisaNet

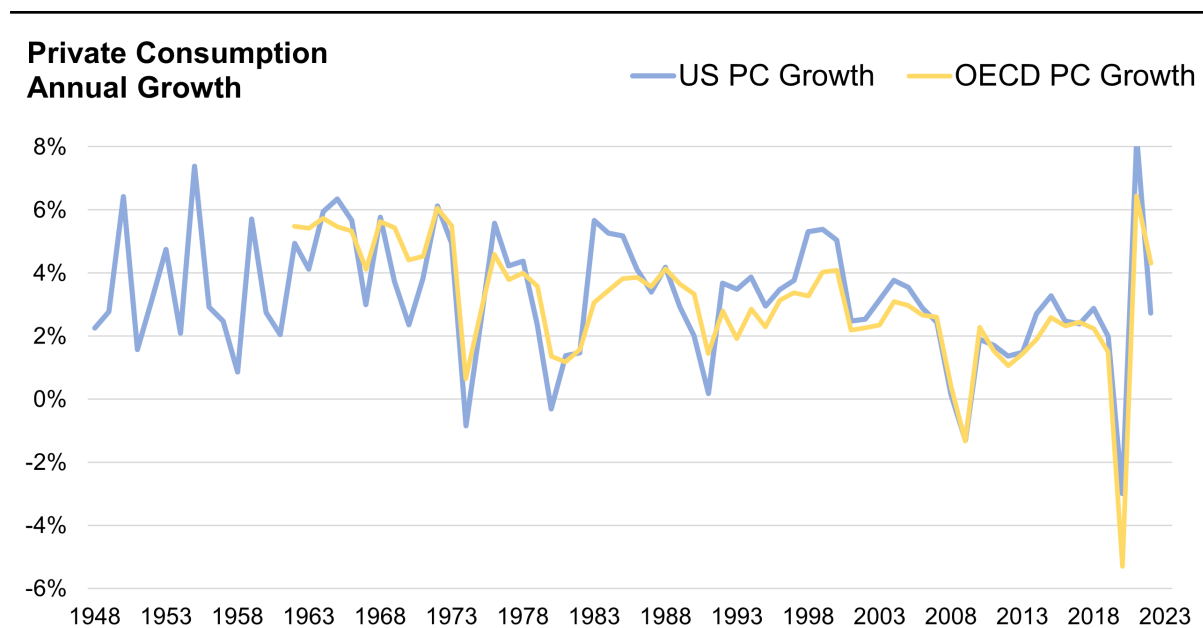
selected emerging markets[17] such as Brazil and India. As can be seen in Figure (A.3), there is a positive correlation between Visa's revenue and private consumption in the core US and EU markets. With all OECD countries' private consumption, the correlation with Visa's revenue growth has been 0.76 in the period from 2009 to 2022. However, also note the revenue growth of Visa has consistently been higher than the private consumption growth in developed markets.

Figure A.3: Year-on-Year (YoY) quarterly growth in Visa's revenue and US and EU private consumption



Source: Bloomberg & OECD

Figure A.4: Annual growth in the US and OECD private consumption



Source: OECD

In addition to OCED countries, emerging markets especially in the Asia-Pacific and partly also South America regions are demanding digital payments. The global market for digital payments in the next 5

years is estimated to grow with a CAGR of 13 %, while for emerging markets it is estimated to be 25 % [30]. However, there are political concerns that make expansion into some emerging markets difficult which is covered in point 4.

2. Electronic and Cross-border Payments Increase

The growth of the payment network market is both decided by private consumption and also the share of that private consumption happening electronically. This is why Visa has experienced long-term revenue growth rates way above private consumption growth. While it may appear in Denmark that all developed markets are far in the transition from cash to electronic payments, this is not the case. In the US, the Federal Reserve Payments Study (2022) [19] estimates only approximately 46 % of private consumption is card payments³. The transaction volume of card payments has grown at a CAGR of 10 % in the period 2018-2021 as this share has increased. In large parts of Europe, private consumption is still highly dependent on cash transactions and Germany is estimated by the Bundes Bank (2022) [20] to only have 42 % of transactions happening electronically⁴. In addition, with increased online retailing across borders and more frequent international travel, the share of international payments is expected to increase which is lucrative for Visa as outlined in the company presentation in Section 2.

3. Political: Regulation against anti-competitive practices and pricing power

Visa and its main competitor MasterCard have been under political scrutiny in its core markets by both US and European regulators in addition to its own business partners, specifically the stores cooperating with Visa and MasterCard. The two main themes for the political criticism have been *i*) Visa's product is critical infrastructure in financial services and therefore should be regulated *ii*) anti-competitive practices against retailers. While the themes certainly overlap, we have made the division into the two most financially relevant topics of regulators' scrutiny of Visa and Visa's litigation for anti-competitive practices allegations in civil court systems.

Despite Visa emphasising its view of itself as a software company, Visa has a difficult time shaking the regulatory scrutiny of being a financial services player providing critical infrastructure due to its importance in the financial system in both home markets and when potentially entering new countries. For the US market, the most notable recent legislation is the Credit Card Competition Act of 2022 which would force card issuers to offer two or more independent payment networks to use ⁵. However, the bill is by political analysts widely not considered to pass and has not even seen action in committees since July 2022. In Europe, the European Commission introduced the harshest legislation with Antitrust: Regulation on Interchange Fees (IFR) which most notably put a price cap on interchange fees (IFR) at 0.2% for debit cards and 0.3% for credit cards. However, since the directive went into effect in 2015, Visa (and MasterCard) have not reduced their fees and instead, other players in the payment infrastructure system have reduced their fees to accommodate the legislation ⁶. Overall, our assessment is regulation will not reduce Visa's revenue in home markets by either a price cap or by forcing additional competition. For emerging markets where Visa is looking to enter, major countries have with increasing effort set up national solutions as their domestic payment network, especially in periods of political turmoil with

³Calculated as total card payment volume divided by the total private expenditures

⁴Bundes Bank (2022) state 58 % is cash transactions meaning only 42 % must be electronic transactions

⁵<https://www.congress.gov/bills/117/congress/senate/bills/4674>

⁶https://ec.europa.eu/commission/presscorner/detail/fr/MEMO_16_2162

the US. From Visa's 2022 annual report [5], they state "Certain governments, including China, India, Indonesia, Thailand and Vietnam, have taken actions to promote domestic payments systems [...] by imposing regulations that favor domestic providers." While China is not a realistic target with recent political tension with the US, both Thailand and India are likely growth prospects for Visa, but mostly within international transactions. Our assessment is Visa will continue to get a foothold in these markets especially within cross-border purchases.

4. Innovations in the FinTech and Payment Solutions Space

The FinTech and Payment Solutions Space is fast-moving with thousands of startups [6] in recent years due to the highly attractive market growth as covered in drivers 1 and 2. While it creates the threat of substitute products which is covered in Porter's analysis, it also creates opportunities as the efficiency of payment solutions can continue to increase. In Visa's core C2B market, Visa has either developed or incorporated new successful payment solutions:

1. Tap to Pay: tap the card on payment terminals
2. Tokenization: transferring payment details to a 16-digit code such that card details are never given to merchants and acquirers which provides an additional layer of security.
3. Click to Pay: connecting a payment card to a phone adds convenience to online card payments and an extra layer of security from the phone (such as face-id or a password).

Especially Tap to Pay has been a major growth driver for Visa as it has been able to charge higher fees to make the feature available on issued payment cards. Overall, continued new technological advances raise the stakes of being a player in the payment solutions market, as some players will likely become obsolete while others will be able to take increase revenue growth with new technological value created.

5. Legal: Litigation against anti-competitive practices and pricing power

Visa has and continues to face numerous lawsuits for abusing its dominant market position and charging inflated fees for stores to use their payment solutions as witnessed by their 861 million dollar expenses for litigation in 2022 [Sustainalytics]. Two of the most significant and recent examples are a 5.6 billion settlement for Visa and MasterCard with US retailers upheld and to be paid in 2023 ⁷ and a similar case in the making with UK retailers ⁸. While Visa claims (and appears) to have stopped the anti-competitive practices that it was held liable for in the US class action suit, there is still litigation in the pipeline against Visa for at least 3-5 years. In addition, the positive results for retailers so far have enticed additional retailers to seek civil court compensation from Visa. Therefore, litigation expenses are likely to remain elevated during the foreseeable future.

<https://usa.visa.com/visa-everywhere/global-impact/moodys-analytics-study-the-global-impact-of-electronic-payments.html>

<https://usa.visa.com/partner-with-us/visa-consulting-analytics/economic-insights/the-plot-thickens-for-2023.html>

⁷<https://www.reuters.com/legal/visa-mastercard-56-bln-settlement-with-retailers-is-upheld-2023-03-15/>

⁸<https://news.sky.com/story/mastercard-and-visa-to-be-hit-with-7-5bn-compensation-claim-12816031>

Cash flow statement

The cash flow statement was reformulated by only moving a few items to different sections, again it is important to look at apples to apples.

A.3 Reported Financial statements

The source for all financial statements is Visa, Inc.

Reported Balance sheets						
\$USD Million						
	2017	2018	2019	2020	2021	2022
Cash and Cash equivalents	9.874	8.162	7.838	16.289	16.487	15.689
Restricted cash equivalents	1.031	1.491	1.205	901	894	1.449
Investment Securities	3.564	3.547	4.236	3.752	2.025	2.833
Settlement receivable	1.422	1.582	3.048	1.264	1.758	1.932
Accounts receivable	1.132	1.208	1.562	1.618	1.968	2.020
Customer collateral	1.106	1.324	1.648	1.850	2.260	2.342
Current Portion of Client incentives	344	340	741	1.214	1.359	1.272
Prepaid expenses and other current assets	550	562	712	757	856	2.668
Total Current assets	19.023	18.216	20.970	27.645	27.607	30.205
Investment Securities	1.926	4.082	2.157	231	1.705	2.136
Client Incentives	591	538	2.084	3.175	3.245	3.348
Property Equipment and Technology	2.253	2.472	2.695	2.737	2.715	3.223
Goodwill	15.110	15.194	15.656	15.910	15.958	17.787
Intangible assets, net	27.848	27.558	26.780	27.808	27.664	25.065
Other assets	1.226	1.165	2.232	3.413	4.002	3.737
Total assets	67.977	69.225	72.574	80.919	82.896	85.501
Accounts payable	179	183	156	174	266	340
Settlement payable	2.003	2.168	3.990	1.736	2.443	3.281
Customer collateral	1.106	1.325	1.648	1.850	2.260	2.342
Accrued compensation and benefits	757	901	796	821	1.211	1.359
Client incentives	2.089	2.834	3.997	4.176	5.243	6.099
Accrued liabilities	1.129	1.160	1.625	1.840	2.334	3.726
Current maturities of debt	1.749	-	-	2.999	999	2.250
Deferred purchase consideration	-	1.300	-	-	-	-
Accrued litigation	982	1.434	1.203	914	983	1.456
Total Current liabilities	9.994	11.305	13.415	14.510	15.739	20.853
Long Term Debt	16.618	16.630	16.729	21.071	19.978	20.200
Deferred tax liabilities	5.980	4.618	4.807	5.237	6.128	5.332
Other Non-Current Liabilities	2.625	2.666	2.939	3.891	3.462	3.535
Total Long Term Liabilities	25.223	23.914	24.475	30.199	29.568	29.067
Total Liabilities	35.217	35.219	37.890	44.709	45.307	49.920
Preferred stock	5.526	5.470	5.462	5.086	3.080	2.324
Common stock	16.900	16.678	16.541	16.721	18.855	19.545
Retained Earnings	9.508	11.318	13.502	14.088	15.351	16.116
Comprehensive Income	878	547	650	354	436	2.369
Other Share Holders Equity	- 52	- 7	- 171	- 39	- 133	- 35
Share Holder Equity	32.760	34.006	34.684	36.210	37.589	35.581
Total Liabilities And Equity	67.977	69.225	72.574	80.919	82.896	85.501

Reported Comprehensive Income Statement						
USD Million						
	2017	2018	2019	2020	2021	2022
Net Revenue	18.358	20.609	22.977	21.846	24.105	29.310
Operating Expenses						
Personnel	2.628	3.170	3.444	3.785	4.240	4.990
Marketing	922	988	1.105	971	1.136	1.336
Network and processing	620	686	721	727	730	743
Professional fees	409	446	454	408	403	505
Depreciation and amortization	556	613	656	767	804	861
General and administrative	1.060	1.145	1.196	1.096	985	1.194
Litigation provision	19	607	400	11	3	868
Total operating expenses	6.214	7.655	7.976	7.765	8.301	10.497
Operating Income	12.144	12.954	15.001	14.081	15.804	18.813
Non-operating Income (Expense)						
Interest expense	- 563	- 612	- 533	- 516	- 513	- 538
Investment Income (expense) and other	113	464	416	225	772	139
Total non-operating income (expense)	- 450	- 148	- 117	- 291	- 259	- 677
Income before income taxes	11.694	12.806	14.884	13.790	16.063	18.136
Income tax provision	4.995	2.505	2.804	2.924	3.752	3.179
Net income	6.699	10.301	12.080	10.866	12.311	14.957
Other comprehensive income (loss) all after tax						
Investment securities	37	90	16	3	4	105
Defined benefit pension	149	15	131	9	147	120
Derivative instruments	14	96	139	490	34	675
Foreign currency translation adjustments	1.136	352	1.228	1.511	95	3.255
Total other comprehensive income	1.336	331	1.204	1.027	82	2.805
Total Comprehensive income	8.035	9.970	10.876	11.893	12.393	12.152

Reformulated Cash flow Statement						
\$USD Million						
	2017	2018	2019	2020	2021	2022
Operating Activities						
Net Income/Loss	6.699	10.301	12.080	10.866	12.311	14.957
Total Depreciation And Amortization	556	602	656	767	804	861
Other Non-Cash Items	6.537	4.477	6.351	7.205	8.814	10.688
<i>Total Non-cash items</i>	<i>7.093</i>	<i>5.079</i>	<i>7.007</i>	<i>7.972</i>	<i>9.618</i>	<i>11.549</i>
Change In Accounts Receivable	- 54 -	70 -	333 -	43 -	343 -	97
Change In Accounts Payable	- 30	3 -	24	21	88	67
Change in Assets/Liabilities	- 4.857 -	4.584 -	6.342 -	9.009 -	7.446 -	9.158
<i>Total Change in Assets/Liabilities</i>	<i>- 4.475 -</i>	<i>2.439 -</i>	<i>6.303 -</i>	<i>8.398 -</i>	<i>6.702 -</i>	<i>7.657</i>
Cash Flow From Operating Activities	9.317	12.941	12.784	10.440	15.227	18.849
Investing Activities						
Net Property, Plant, And Equipment	- 695 -	718 -	756 -	736 -	705 -	970
Net Change In Intangible Assets						
Net Acquisitions/Divestitures	- 302 -	196 -	699 -	77 -	75 -	1.948
Cash Flow From Investing Activities	- 997 -	914 -	1.455 -	813 -	780 -	2.918
Financial Activities						
Net Common Equity Issued/Repurchased	- 6.742 -	7.028 -	8.445 -	7.924 -	8.468 -	11.393
Repayments of debt	0	-1750	0	0	-3000	-1000
Dividends payed	-1579	-1918	-2269	-2664	-2798	-3203
Proceeds from issuance of debt	2488	0	0	7212	0	3218
Financial Activities - Other	- 91 -	94 -	1.347 -	592 -	144 -	318
Net Change in Short-term Investments	4					
Net Change in Long-Term Investments	1.728 -	2.186	842	2.168	519 -	1.498
Investing Activities - Other		16	22	72	109	128
Cash Flow From Financial Activities	- 4.192 -	12.960 -	11.197 -	1.728 -	13.782 -	14.066
Net Cash Flow	4.364 -	1.034 -	145	8.339	628	578
Start of year	7.647	12.011	10.977	10.832	19.171	19.799
End of year	12.011	10.977	10.832	19.171	19799	20.377

		2017	2018	2019	2020	2021	2022
Primo		27.195	27.234	28.536	29.222	31.124	34.509
Comprehensive income		8.035	9.970	10.876	11.895	12.393	12.152
VE territory covered losses	-	209 -	11 -	172 -	37 -	147 -	43
New accounting standards		-	-	392	-	3	-
Right to recover for Covered losses		191	56	8			
Recovery through conversion rate		-	-	-	169	53	141
Issuance of series A preferred stock		-	-	-	-		-
Conversion of Common stock					207	1.951	612
Charitable contribution of shares		170	-	-	-	-	-
Treasury stock appreciation		14	-	-	-	-	-
Share-based compensation		235	327	407	416	542	602
Stock Issued under equity plans		-	-	-	190	208	196
Restricted stock	-	76 -	94 -	111 -	160 -	144 -	120
Cash from employee equity plans		149	164	162	-	-	-
Cash dividends paid quaterly	-	1.579 -	1.918 -	2.269 -	2.664 -	2.798 -	3.203
Repurchase of common stock	-	6.891 -	7.192 -	8.607 -	8.114 -	8.676 -	11.589
Ultimo		27.234	28.536	29.222	31.124	34.509	33.257

A.4 Reformulated Financial statements

Reformulated Cash flow Statement						
\$USD Million						
	2017	2018	2019	2020	2021	2022
Operating Activities						
Net Income/Loss	6.699	10.301	12.080	10.866	12.311	14.957
Total Depreciation And Amortization	556	602	656	767	804	861
Other Non-Cash Items	6.537	4.477	6.351	7.205	8.814	10.688
<i>Total Non-cash items</i>	<i>7.093</i>	<i>5.079</i>	<i>7.007</i>	<i>7.972</i>	<i>9.618</i>	<i>11.549</i>
Change In Accounts Receivable	- 54 -	70 -	333 -	43 -	343 -	97
Change In Accounts Payable	- 30	3 -	24	21	88	67
Change in Assets/Liabilities	- 4.857 -	4.584 -	6.342 -	9.009 -	7.446 -	9.158
<i>Total Change in Assets/Liabilities</i>	<i>- 4.475 -</i>	<i>2.439 -</i>	<i>6.303 -</i>	<i>8.398 -</i>	<i>6.702 -</i>	<i>7.657</i>
Cash Flow From Operating Activities	9.317	12.941	12.784	10.440	15.227	18.849
Investing Activities						
Net Property, Plant, And Equipment	- 695 -	718 -	756 -	736 -	705 -	970
Net Change In Intangible Assets						
Net Acquisitions/Divestitures	- 302 -	196 -	699 -	77 -	75 -	1.948
Cash Flow From Investing Activities	- 997 -	914 -	1.455 -	813 -	780 -	2.918
Financial Activities						
Net Common Equity Issued/Repurchased	- 6.742 -	7.028 -	8.445 -	7.924 -	8.468 -	11.393
Repayments of debt	0	-1750	0	0	-3000	-1000
Dividends payed	-1579	-1918	-2269	-2664	-2798	-3203
Proceeds from issuance of debt	2488	0	0	7212	0	3218
Financial Activities - Other	- 91 -	94 -	1.347 -	592 -	144 -	318
Net Change in Short-term Investments	4					
Net Change in Long-Term Investments	1.728 -	2.186	842	2.168	519 -	1.498
Investing Activities - Other		16	22	72	109	128
Cash Flow From Financial Activities	- 4.192 -	12.960 -	11.197 -	1.728 -	13.782 -	14.066
Net Cash Flow	4.364 -	1.034 -	145	8.339	628	578
Start of year	7.647	12.011	10.977	10.832	19.171	19.799
End of year	12.011	10.977	10.832	19.171	19799	20.377

Reformulated CORE Comprehensive Income Statement						
\$USD Million						
	2017	2018	2019	2020	2021	2022
Net Revenue	17.517	19.665	21.664	20.414	22.430	27.319
Operating Expenses						
Network and processing	- 620	- 686	- 721	- 727	- 730	- 743
Personnel	- 2.628	- 3.170	- 3.444	- 3.785	- 4.240	- 4.990
Marketing	- 922	- 988	- 1.105	- 971	- 1.136	- 1.336
Professional fees	- 409	- 446	- 454	- 408	- 403	- 505
Depreciation and amortization	- 556	- 613	- 656	- 767	- 804	- 861
General and administrative	- 1.060	- 1.145	- 1.196	- 1.096	- 985	- 1.194
Litigation provision	- 19	- 607	- 400	- 11	- 3	- 868
Operating income from sales (Before tax)	11.303	12.010	13.688	12.649	14.129	16.822
Tax on operating income from sales						
Tax as reported	4995	2505	2804	2924	3752	3179
Tax on other operating income	- 294	- 198	- 276	- 301	- 352	- 418
Tax on financial income	197	129	112	108	108	113
Tax on unusual items	- 40	- 97	- 87	- 47	- 162	- 29
Tax liability	4858,15	2337,84	2552,84	2684,39	3345,86	2903,06
(CORE) Operating Income sales after tax	6.445	9.672	11.135	9.965	10.783	13.919
Other operating income						
Other Operating Income Before tax	841	944	1.313	1.432	1.675	1.991
Tax effect	- 294	- 198	- 276	- 301	- 352	- 418
Net other operating income	547	746	1.037	1.131	1.323	1.573
Total CORE Operating income	6.992	10.418	12.172	11.096	12.106	15.492
Unusual Items						
Investment Income/expense and other	113	464	416	225	772	- 139
Tax effect Unusual items	- 40	- 97	- 87	- 47	- 162	- 29
Investment securities	37	-90	16	-3	-4	-105
Derivative instruments	14	96	139	-488	34	675
Foreign currency translation adjustments	1136	-352	-1228	1511	-95	-3255
Defined benefit pension	149	15	131	9	147	- 120
Total unusual items	1.409	36	875	1.207	692	2.915
Financial activities						
Interest expense net	- 563	- 612	- 533	- 516	- 513	- 538
Tax shield	197	129	112	108	108	113
Net financial costs/gains after tax	- 366	- 483	- 421	- 408	- 405	- 425
Total Comprehensive Income after tax	8.035	9.970	10.876	11.895	12.393	12.152

Reformulated Balance sheets						
\$USD Million						
(In \$USD Millions)	2017	2018	2019	2020	2021	2022
Operating Assets			1.466 -	1.784	494	174
Settlement receivable	1.422	1.582	3.048	1.264	1.758	1.932
Customer collateral	1.106	1.324	1.648	1.850	2.260	2.342
Current Portion of Client incentives	344	340	741	1.214	1.359	1.272
Prepaid expenses and other current assets	550	562	712	757	856	2.668
Client Incentives	591	538	2.084	3.175	3.245	3.348
Property Equipment and Technology	2.253	2.472	2.695	2.737	2.715	3.223
Goodwill	15.110	15.194	15.656	15.910	15.958	17.787
Intangible assets, net	27.848	27.558	26.780	27.808	27.664	25.065
Other assets	1.226	1.165	2.232	3.413	4.002	3.737
Accounts receivable	1.132,0	1.208,0	1.562,0	1.618,0	1.968,0	2.020,0
Total Operating assets	51.582	51.943	57.158	59.746	61.785	63.394
Operating Liabilities						
Settlement payable	2.003	2.168	3.990	1.736	2.443	3.281
Customer collateral	1.106	1.325	1.648	1.850	2.260	2.342
Accrued compensation and benefits	757	901	796	821	1.211	1.359
Client incentives	2.089	2.834	3.997	4.176	5.243	6.099
Accrued liabilities	1.129	1.160	1.625	1.840	2.334	3.726
Accrued litigation	982	1.434	1.203	914	983	1.456
Accounts payable	179	183	156	174	266	340
Deferred tax liabilities	5.980	4.618	4.807	5237	6.128	5.332
Other Non-Current Liabilities	2.625	2.666	2.939	3.891	3.462	3.535
Total Operating liabilities	16.850	17.289	21.161	20.639	24.330	27.470
NOA	34.732	34.654	35.997	39.107	37.455	35.924
Financial Assets						
Cash and Cash equivalents	9.874,0	8.162,0	7.818,0	16.289,0	16.487,0	15.689,0
Restricted cash equivalents	1.031,0	1.491,0	1.205,0	901,0	894,0	1.449,0
Investment Securities	3.564,0	3.547,0	4.236,0	3.752,0	2.025,0	2.833,0
Investment Securities	1.926,0	4.082,0	2.157,0	231,0	1.705,0	2.136,0
Total Financial Assets	16.395,0	17.282,0	15.416,0	21.173,0	21.111,0	22.107,0
Financial Liabilities						
Current maturities of debt	1749	0	0	2999	999	2250
Deferred purchase consideration	0	1300	0	0	0	0
Long Term Debt	16.618	16.630	16.729	21.071	19.978	20.200
Preferred stock	5.526	5.470	5.462	5.086	3.080	2.324
Total Financial Liabilities	23.893	23.400	22.191	29.156	24.057	24.774

NFA	-	7.498	-	6.118	-	6.775	-	7.983	-	2.946	-	2.667
Common stock		16.900		16.678		16.541		16.721		18.855		19.545
Retained Earnings		9.508		11.318		13.502		14.088		15.351		16.116
Comprehensive Income		878		547	-	650		354		436	-	2.369
Other Share Holders Equity	-	52	-	7	-	171	-	39	-	133	-	35
Share Holder Equity		27.234		28.536		29.222		31.124		34.509		33.257

Reformulated comprehensive

A.5 Reformulation of the financial statements

Changes in equity statement

The first part of the financial statement that will be reformulated is the consolidated statements of changes in equity. As highlighted by Penman (2013) it is important to ensure that all comprehensive income is captured in the equity statement. This will ensure that accounting relations used later in the analysis will hold. Furthermore, it is important to keep an eye open for any preferred shareholders. As we wish to create a valuation of the common shareholder's equity any dividends to preferred shareholders or preferred shareholder equity should effectively be moved to either expenses or debt, respectively. This is due to the order of repayment in case of a default where the preferred shareholders are further ahead in the line compared to common shareholders.

Balance

The balance sheet is a financial statement that provides a snapshot of a company's financial position at a specific point in time. The reformulation of the balance is the first real step to separate Visa's operations from its investing and financial activities. The main result of the reformulation is to identify all of Visa's operating assets and liabilities, and their financial assets and liabilities. Investing activities will thus be split into either operating or financial activities and some activities might be reclassified as either operating or financial. The rule of thumb is to see if the activity is a placement of excess cash which thus will be a financial activity. Furthermore, as Penman (2013) says all mark-to-market fair value measured balance posts is categorized as financial assets/liabilities.

Noticeable re-classifications include:

Cash and equivalents are classified as financial assets. As Visa does not report the amount of cash needed for operations we follow Penman(2013) and classify all cash and equivalents as financial assets.

Investment Securities are all registered as financial assets. We do not consider any of these as strategical investments other than placement of excess cash. Furthermore, these are measured as mark-to market and should thus be considered financial assets, Penman (2013). This is both the current and non-current assets in investment securities that are classified as financial assets.

Current maturities of debt, long term debt, and Deferred purchase consideration are all classified as a financial assets as their mark-to-market fair value is reported in the balance sheet.

Preferred shareholder equity is also classified as a financial liabilities as we wish to analyze the common shareholders equity.

Statement of operations

The statement of operations is an overview of the revenues and expenses for Visa. The reformulated statement of operations wishes to show the income, expenses, and tax burdens to the following parts of

the business: core operating income, other operating income, and financial activities.

As explained in section ?? it is important to use all comprehensive income and thus the statement of operations is combined with the statement of comprehensive income.

The reformulation of the balance sheet allows us to correctly identify which revenues and expenses are operational, or financial. It is very important to compare apples to apples, Penman (2013).

The tax rates used for the period is noticeably different as the US changed their corporate tax level from 2017 to 2018 where the former tax rate was at 35 pct. and was changed to 21 pct. from 2018 onward.

The income statement is next cleaned for unusual items. A large part of Visa's business is not registered in USD as the financial report is and thus, there are large differences in currency translations across business areas.

A.6 Bonds in Cost of Debt

Visa Bonds	MATURITY	ISIN	COUPON	Yield to Maturity	Included?
Visa 2050	01/01/2050	US92826CAQ50	2,00%	4,70%	Included
Visa 2047	15/09/2047	US92826CAJ18	3,65%	4,72%	Included
Visa 2045	14/12/2045	US92826CAF95	4,30%	4,75%	Included
Visa 2040	15/04/2040	US92826CAK80	2,70%	4,39%	Included
Visa 2035	14/12/2035	US92826CAE21	4,15%	4,18%	Included
Visa 2030	15/04/2030	US92826CAM47	2,05%	4,17%	Included
Visa 2027	15/09/2027	US92826CAH51	2,75%	4,19%	Included
Visa 2026	15/09/2026	XS2479941499	1,50%	3,53%	NOT INCLUDED
Visa 2025	14/12/2025	US92826CAD48	3,15%	4,44%	Included

US Bonds	MATURITY	ISIN	COUPON	Yield to Maturity	Included?
US30Y	25/04/2053			3,65%	Included
US20Y	25/04/2043			3,77%	Included
US10Y	25/04/2033			3,40%	Included
US7Y	25/04/2030			3,42%	Included
US5Y	25/04/2028			3,43%	Included
US2Y	25/04/2025			3,86%	Included
US1Y	25/04/2024			4,60%	Included
US6M	25/10/2023			4,98%	Included

A.7 Corporate Credit Spreads

Figure A.5: Corporate Credit Spreads

$>$	$\leq to$	Rating is	Spread is
-100000	0.199999	D2/D	20.00%
0.2	0.649999	C2/C	17.50%
0.65	0.799999	Ca2/CC	15.78%
0.8	1.249999	Caa/CCC	11.57%
1.25	1.499999	B3/B-	7.37%
1.5	1.749999	B2/B	5.26%
1.75	1.999999	B1/B+	4.55%
2	2.2499999	Ba2/BB	3.13%
2.25	2.49999	Ba1/BB+	2.42%
2.5	2.999999	Baa2/BBB	2.00%
3	4.249999	A3/A-	1.62%
4.25	5.499999	A2/A	1.42%
5.5	6.499999	A1/A+	1.23%
6.5	8.499999	Aa2/AA	0.85%
8.50	100000	Aaa/AAA	0.69%

Source: Damodaran

A.8 Fama-French risk premiums

The risk premiums attached to the additional fama-french risk factors are found by the estimation method suggested by Fama & French (2015) [43] which simply is the observed average return differences of portfolios of stocks with the different characteristics. Formally, $\hat{\gamma}_{SMB} = \sum_{t=1}^T f_{t,SMB}$. We use a time horizon of 35 years to be consistent with the historical ERP estimation.

Table A.1: Annual Fama-French Risk Premiums: 03/2023 - 07/1963

Risk premium	$\mathbb{E}[r_m] - r_f$ (ERP)	$\hat{\gamma}_{SMB}$	$\hat{\gamma}_{HML}$	$\hat{\gamma}_{CMA}$	$\hat{\gamma}_{RMW}$
Annual size	4.88 %	2.79 %	3.71 %	3.39 %	3.56 %

Source: Damodaran Implied ERP for April 2023 & Own Calculations on French data [15]

A.9 Theoretical and empirical issues with CAPM

Because of CAPM's simplicity, easily implemented estimations methods, and lack of better alternatives⁹ it is the workhorse for practitioners for determining the cost of equity [33]. However, the model's theoretical framework, underlying assumptions¹⁰, expected return predictions and the most common econometric implementation methods for estimating the model have been widely scrutinized. To motivate other CoE models, we will here briefly mention two issues that could be relevant in determining the CoE for Visa. First, the CAPM is a static model meaning β_i is defined as a constant for Visa. Both practitioners and academics widely accept that this is not true e.g. Fama and MacBeth (1973) show while β_i and returns are correlated, the relationship is unpredictable from one 5-year period to the next [53] <https://www.jstor.org/stable/1831028>. Second, research from 1970 to today has found that realized stock returns deviate systematically from what is predicted in the CAPM. Black, Jensen, and Scholes (1972) [BJS1972] found a positive correlation between stock returns and their beta supporting the CAPM [53]. However, the relationship deviated systematically from what was predicted in the CAPM with low-beta stocks having a too-high expected return and high-beta stocks having a too low expected return (the security market line was too flat). Fama & French (1992) showed returns could be predicted by both the size and book-to-market value of companies. Since has hundreds of other risk premiums shown to be statically significant to predict returns with varying degree of economic reasoning [44].

A.10 DCF: Present Vale of Forecasted Cash Flows 2023 - 2033

Figure A.6: Financial performance 2018 to 2022

	REPORTED				FORECAST								
	30/09/2022	30/09/2023	30/09/2024	30/09/2025	30/09/2026	30/09/2027	30/09/2028	30/09/2029	30/09/2030	30/09/2031	30/09/2032	Terminal år	
Net revenue	27.319	30.626	33.942	37.319	40.816	44.445	48.185	52.008	55.886	59.786	63.672	65.264,30	
Oil after tax	13.919	14.469	16.036	17.631	19.283	20.998	22.765	24.571	26.403	28.246	30.082	30.834	
Other Oil after tax	1.573	1.437	1.592	1.751	1.915	2.085	2.261	2.440	2.622	2.805	2.987	3.062	
Total Oil After tax	15.492	15.906	17.628	19.382	21.198	23.083	25.026	27.011	29.026	31.051	33.069	33.896	
Sales Growth		12,11%	10,83%	9,95%	9,37%	8,89%	8,41%	7,94%	7,46%	6,98%	6,50%	2,50%	
Profit margin CORE BUSINESS	51%	47%	47%	47%	47%	47%	47%	47%	47%	47%	47%	47%	
Profit margin other Oil	6%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	
Net operating assets (Invested Capital)	35.924	39.933	42.926	45.777	48.562	51.290	53.933	56.462	58.848	61.062	63.076	64.653	
ΔNOA	1.531	4.009	2.993	2.852	2.784	2.728	2.643	2.529	2.386	2.214	2.014	1.577	
Asset turnover	74,46%	77%	79%	82%	84%	87%	89%	92%	95%	98%	101%	101%	
Cost of capital		9,2%	9,5%	9,3%	9,1%	8,0%	8,0%	8,0%	8,0%	8,0%	8,0%	8,0%	
1+WACC		109,2%	109,5%	109,3%	109,1%	108,0%	108,0%	108,0%	108,0%	108,0%	108,0%	108,0%	
Accumulated discount rate		1,09	1,20	1,31	1,42	1,47	1,59	1,71	1,85	2,00	2,16		
Discount factor		0,92	0,83	0,77	0,71	0,68	0,63	0,58	0,54	0,50	0,46		
FCF		11.897	14.635	16.531	18.414	20.355	22.382	24.482	26.640	28.837	31.056		
Terminal value												578.764	
PV FCF		134.204											
PV Terminal period		268.080											
Enterprise value		402.284											
Net debt		2.667											
Equity value to common shareholders		399.617											
Outstanding shares	1.883.494.671												
Value per share USD		212,17											
Close value 30/09-2022		177,65											
Value 8/5-2023		231,78											

Note: DKKm

Source: Annual reports

A.11 Valuation DCF and Residual income

⁹We will present and use alternative Cost of Equity models, but their advantage over CAPM has been questioned

¹⁰CAPM can be derived using different assumptions, but normal returns, diversified investors, perfect information, and borrowing at the risk-free rate are some of the most scrutinized <https://www.jstor.org/stable/10.1086/499143?seq=1>

		REPORTED	FORECAST										
			2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
Net revenue		27.319	30.626	33.942	37.319	40.816	44.445	48.185	52.008	55.886	59.786	63.672	66.092,04
	OI after tax	13.919	14.469	16.036	17.631	19.283	20.998	22.765	24.571	26.403	28.246	30.082	31.225
	Other OI after tax	1.573	1.437	1.592	1.751	1.915	2.085	2.261	2.440	2.622	2.805	2.987	3.101
	Total OI After tax	15.492	15.906	17.628	19.382	21.198	23.083	25.026	27.011	29.026	31.051	33.069	34.326
	Sales Growth		12,11%	10,83%	9,95%	9,37%	8,89%	8,41%	7,94%	7,46%	6,98%	6,50%	3,80%
Profit margin CORE BUSINESS		51%	47%	47%	47%	47%	47%	47%	47%	47%	47%	47%	47%
	Profit margin other OI	6%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%	5%
	Net operating assets	35.924	39.933	42.926	45.777	48.562	51.290	53.933	56.462	58.848	61.062	63.076	65.473
	ΔNOA	-1.531	4.009	2.993	2.852	2.784	2.728	2.643	2.529	2.386	2.214	2.014	2.397
	RNOA	41%	44%	44%	45%	46%	48%	49%	50%	51%	53%	54%	54%
Cost of capital			9,5%	9,2%	8,4%	8,3%	8,1%	8,0%	8,0%	8,0%	8,1%	8,2%	8,3%
	1+WACC		109,5%	109,2%	108,4%	108,3%	108,1%	108,0%	108,0%	108,0%	108,1%	108,2%	108,2%
	Accumulated discount rate		1,10	1,19	1,27	1,37	1,48	1,58	1,71	1,85	2,02	2,20411	
	Discount factor		0,91	0,84	0,78	0,73	0,68	0,63	0,58	0,54	0,50	0,45	
	FCF		11.897	14.635	16.531	18.414	20.355	22.382	24.482	26.640	28.837	31.056	31.929
Terminal value												702.799	
PV FCF		134.517											
PV Terminal period		318.858											
RNOA			44,3%	44,1%	45,2%	46,3%	47,5%	48,8%	50,1%	51,4%	52,8%	54,2%	54,4%
Residual Income			12.480	13.971	15.765	17.408	19.134	20.941	22.711	24.517	26.282	28.048	29.063
PV Residual Income		127.210											
PV Terminal Period		290.241											
DCF			Enterprise value	453.375	ReOI	Enterprise Value	453.375						
			Net debt	2.667		Net debt	2667						
			Equity value to common shareholders	450.708		Equity value to common shareholders	450.708						
			Outstanding shares, Million	2.090		Outstanding Shares	2.090						
			Value per share USD	215,65		Value per share USD	215,65						
			Value 25/4-2023	229.54		Value 25/4-2023	229.54						

A.12 Residual Income Valuation

Another way of calculating Visa's equity value is the residual income (ReOI) model. A brief description of the model is that only operating income that exceeds the investors' required earnings¹¹ are creating value for investors. Visa's enterprise value can then be interpreted as the sum of the future residual income and value already in the firm, i.e. net operating assets. The residual income valuation provides the same result as the standard DCF and is calculated as follows:

$$\text{ReOI} = \text{NOA} \times (\text{RNOA} - \text{WACC}) \quad (\text{A.1})$$

Table A.2: Valuation using residual income model, Medium terminal growth and main case WACC

(USD Million)	2023	2024	...	2031	2023	Terminal
Residual Income	12,480	13,971	...	26,282	28,048	29,063
Discount Factor	0.92	0.83	...	0.50	0.46	
Present value FCF	127,210					
Present value Terminal year	290,241					
Net Operating assets	35,924					
Enterprise value	453,375					
Enterprise value	453,375					
Net Financial Obligations (Net Debt)	2,667					
Equity Value Common shareholders	450,708					
Number of Outstanding Shares	2,090,000,000					
Share Price, USD	215.65					

Notes: Calculations are done with a medium growth scenario and main scenario cost of capital. Data in USD million unless stated otherwise. 64 pct. of Enterprise Value is from terminal growth

Source: Own calculations and Visa Financial Statement 2017-2022

The main takeaway from Table A.2 is that both the residual income model and discounted cash flow model yield the same result meaning the models have been calculated correctly.

Looking at Table A.2, the residual income model adds a smaller value to the terminal value which might be preferable for some analysts Penman (2013) [2]. This is because more analysis typically goes into the budgeting period so having a larger share of the value here can be viewed as a more reliable result.

A.13 History: the value Visa creates

Visa was founded in 1958 by Bank of America launching BankAmericard where consumers could pay with electronic funds or credit issued by their own bank [VisaOrigin]. The potential value of such a

¹¹Required earnings is defined as $\text{WACC} \times \text{NOA}$

solution was large because merchants faced costs both with issuing credit to consumers themselves and handling large quantities of cash [6]. In addition, card payments provided a convenience and safety value for consumers[6]. While the potential value of BankAmericard was large, two-sided networks such as payment networks, require widespread adoption by both consumers and businesses to be valuable [6]. Consumers will only use the card if merchants accept it, and merchants will only accept it if consumers wish to pay using the card. This resulted in an aggressive growth strategy of mailing 65.000 unsolicited credit cards with \$300 credit to US household [**VisaOrigin**]. While BankAmericard initially resulted in large financial and brand losses for Bank of America, it gained widespread adoption and was already profitable in 1961 [**VisaOrigin**]. In 1976, the name was changed to Visa which has since grown to be the 10th most valuable public company in the world¹².

¹²Source: Measured by Market Cap in Bloomberg at 25-04-2023